

Families of Random Variables

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Topic Overview

- We will study several useful families of random variables that arise in interesting scenarios in statistics.
- Discrete random variables- Bernoulli, Binomial, Poisson, multinomial, hypergeometric
- Continuous random variables- Gaussian, uniform, exponential, chi-square

Bernoulli Distribution

Definition

- Let X be a random variable whose value is 1 when a coin toss produces a heads, and a 0 otherwise. If p is the probability that the coin toss produces a heads, we have:

$$P(X = 1) = p, P(X = 0) = 1-p$$

- This is called a **Bernoulli pmf** with parameter p – named after Jacob Bernoulli. X is called a Bernoulli random variable.
- Note here: the coin need not be unbiased any longer!

Properties

- $E[X] = p(1) + (1-p)(0) = p.$
- $\text{Var}(X) = p(1-p)^2 + (1-p)(0-p)^2 = p(1-p)$
- What's the median? any $x \in (0,1)$ if $p = 0.5$,
1 if $p > 0.5$, otherwise 0
- What is (are) the mode(s)? $\{0,1\}$ if $p = 0.5$, 1 if $p > 0.5$, otherwise 0
- What is its MGF? $1 - p + pe^t$

Binomial Distribution

Definition

- Let X be a random variable denoting the number of heads in a sequence of n independent coin tosses (or *Bernoulli trials*) having *success probability* (i.e. probability of getting a heads) p .
- Then the pmf of X is given as follows:

$$P(X = i) = C(n, i) p^i (1 - p)^{n-i}$$

- This is called the **binomial pmf** with parameters (n, p) .

Definition

- The pmf of X is given as follows:

$$P(X = i) = C(n, i) p^i (1 - p)^{n-i}$$

$$C(n, i) = \frac{n!}{i!(n-i)!}$$

- Explanation: Consider a sequence of trials with i successes and $n-i$ failures. The probability that this sequence occurs is $p^i(1-p)^{n-i}$, by the *product rule*. But there are $C(n, i)$ such sequences – so we add their individual probabilities using the *sum rule*.

Definition

- Example: In 5 coin tosses, if we had two heads and three tails, the possible sequences are:
HHTTT, HTHTT, HTTHT, HTTTH, TTTHH,
THTTH, TTHHT, TTTHH, THTHT, THHTT
- What's the probability that a sequence of Bernoulli trials produces a success only on the i -th trial? Note that this is **not** a binomial distribution.

Definition

- The pmf of X is given as follows:

$$P(X = i) = C(n, i) p^i (1 - p)^{n-i}$$

$$C(n, i) = \frac{n!}{i!(n-i)!}$$

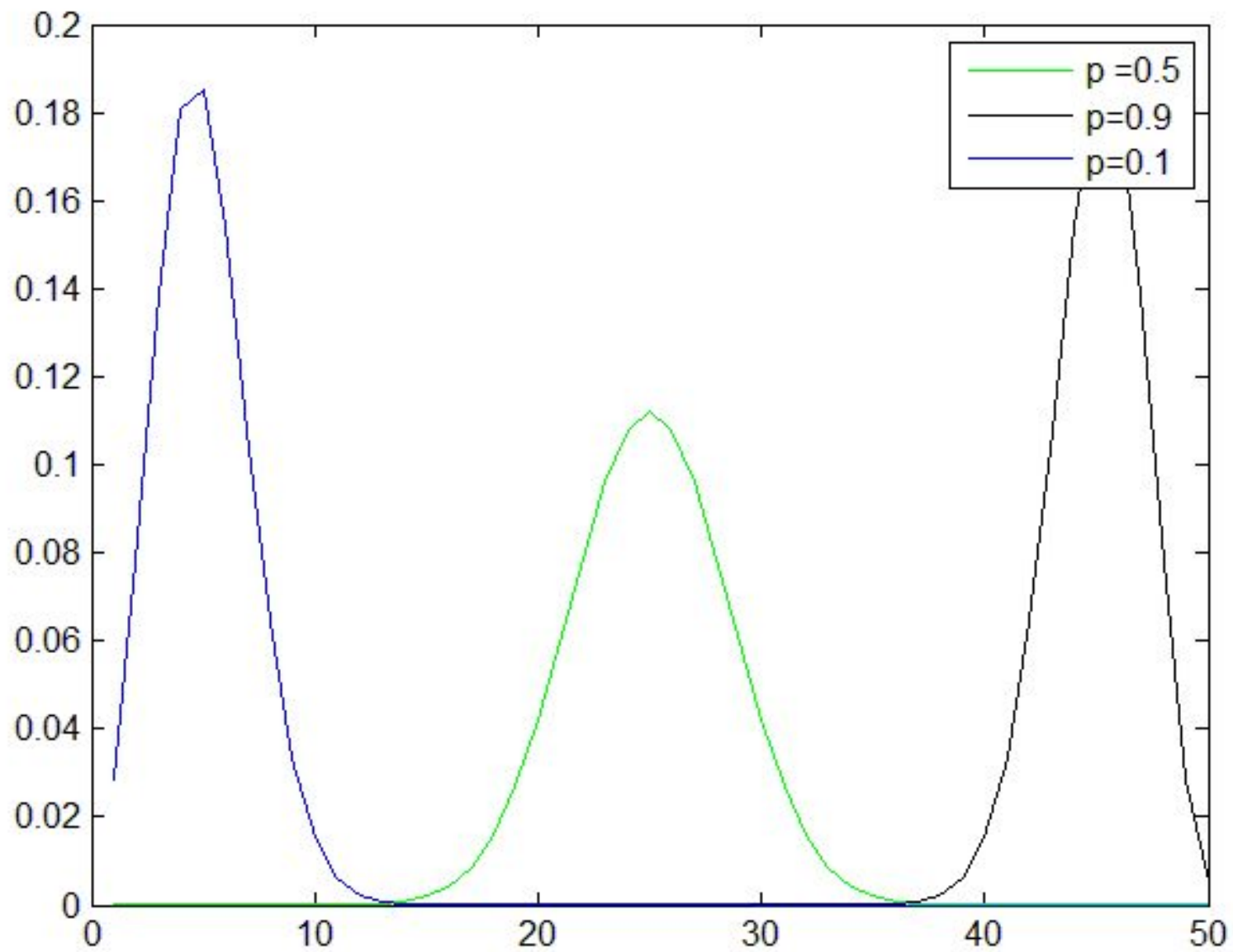
- To verify it is a valid pmf:

$$\sum_{i=1}^n P(X = i) = \sum_{i=1}^n C(n, i) p^i (1 - p)^{n-i}$$

$$= (p + (1 - p))^n$$

← Binomial theorem!

$$= 1$$



Example 1

- Let's say you design a smart self-driving car and you have tested it thoroughly. You have determined that the probability that your car collides is p . You go for an international competition and the rules say that you will win a prize if in k difficult tests, your car collides at most once. What's the probability that you will win the prize?
- Answer: X is the number of times your car collides. X is a binomial random variable with parameters (k, p) . The probability that you win a prize is $P(X \leq 1) = (1-p)^k + C(k, 1)p(1-p)^{k-1}$.

Example 2

- At least half of an airplane's engines need to function for it to operate. If each engine independently fails with probability p , for what values of p is a 4-engine airplane more likely to operate than a 2-engine airplane?
- Answer: The number of functioning engines (X) is a binomial random variable with parameters $(4, p)$ or $(2, p)$.
- In 4-engine case, the probability of operation is $P(X=2) + P(X=3) + P(X=4) = 6p^2(1-p)^2 + 4p(1-p)^3 + (1-p)^4$.
- In 2-engine case, it is $P(X=1) + P(X=2) = 2p(1-p) + (1-p)^2$. Do the math!
- Answer is for $p < 1/3$.

Properties

- Mean: Recall that binomial random variable X is the *sum* of random variables of the following form:

$X_i = 1$ if trial i yields success (this occurs with prob. p)
 $= 0$ otherwise

- Hence

$$E(X) = E\left(\sum_{i=1}^n X_i\right) = np$$

Notice how we are making use of the linearity of the expectation operator. These calculations would have been much harder had you tried to plug in the formulae for binomial distribution.

- Variance:

We are using independence of the X_i values here.

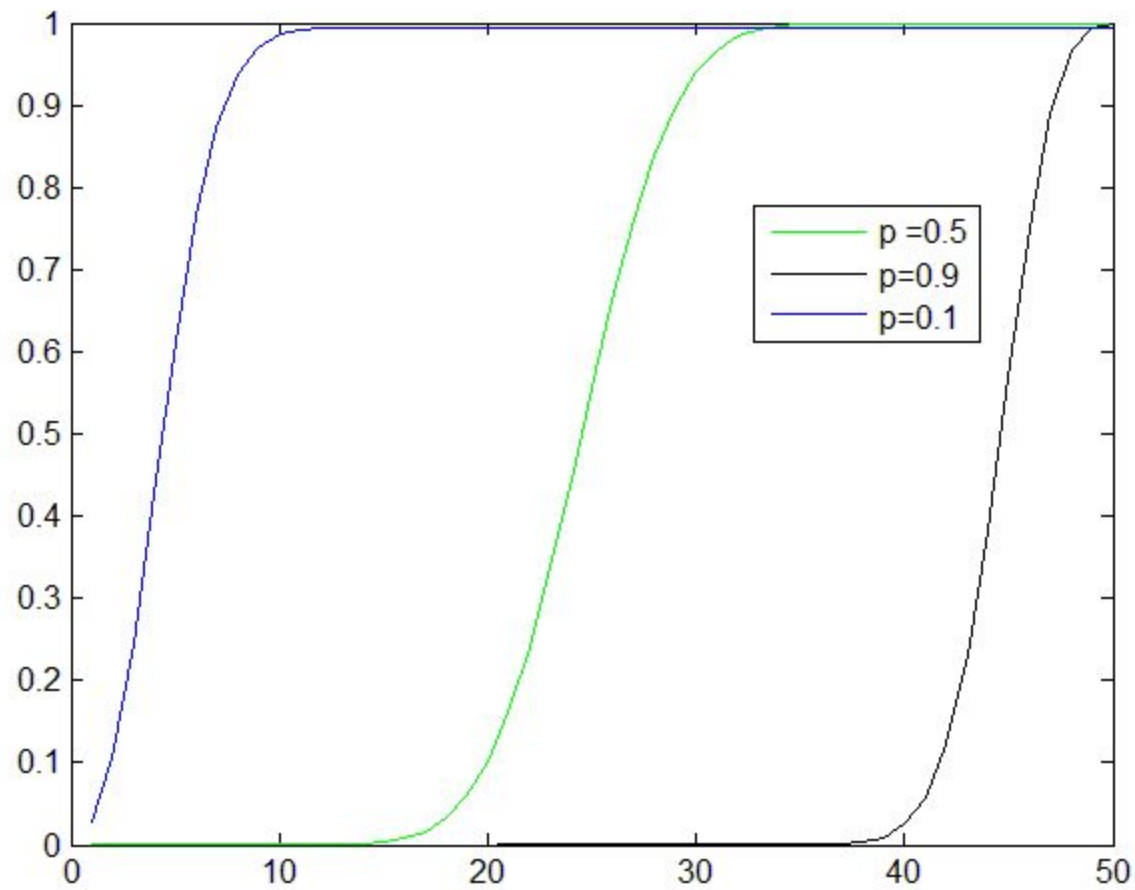
$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) = np(1 - p)$$

Properties

- The CDF is given as follows:

$$F_X(i) = P(X \leq i) = \sum_{k=0}^i C(n, k) p^k (1-p)^{n-k}$$

- MGF: Consider a binomial r.v. to be the sum of n independent Bernoulli trials. Hence using a property of the MGF (which one?), the MGF of the binomial r.v. is given by $(1-p+pe^t)^n$.



Example

- Bits are sent over a communications channel in packets of 12. If the probability of a bit being corrupted over this channel is 0.1 and such errors are independent, **what is the probability that no more than 2 bits in a packet are corrupted?**
- If 6 packets are sent over the channel, what is the probability that **at least one** packet will contain 3 or more corrupted bits?
- Let X denote the number of packets containing 3 or more corrupted bits. What is the probability that X will exceed its mean by more than 2 standard deviations?

Example

- We want $P(X = 0) + P(X=1) + P(X=2)$.
- $P(X=0) = (0.1)^0(0.9)^{12} = 0.282$
- $P(X=1) = C(12,1) (0.1)^1(0.9)^{11} = 0.377$
- $P(X=2) = C(12,2) (0.1)^2(0.9)^{10} = 0.23$
- So the answer is 0.889.

Example

- The probability that 3 or more bits are corrupted is $1 - 0.889 = 0.111$
- Let Y = number of packets with 3 or more corrupted bits. Then we want $P(Y \geq 1) = 1 - P(Y=0)$
 $= 1 - C(6,0)(0.111)^0 (0.889)^6 = 0.4936$.
- Mean of Y is $\mu = 6(0.111) = 0.666$.
- Standard deviation of Y is $\sigma = [6(0.111)(0.889)]^{0.5} = 0.77$.
- We want $P(Y > \mu + 2\sigma) = P(Y > 2.2) = P(Y \geq 3) = 1 - P(Y=0) - P(Y=1) - P(Y=2) = ?$

Example

- We want $P(Y > \mu + 2\sigma) = P(Y > 2.2) = P(Y \geq 3) = 1 - P(Y=0) - P(Y=1) - P(Y=2) = ?$
- $P(Y=0) = C(6,0) (0.111)^0 (0.889)^6 = 0.4936$
- $P(Y=1) = C(6,1) (0.111) (0.889)^5 = 0.37$
- $P(Y=2) = C(6,2) (0.111)^2 (0.889)^4 = 0.115$
- $P(Y > \mu + 2\sigma) = 1 - (0.4936 + 0.37 + 0.115) = 0.0214$

Related distributions

- In a sequence of Bernoulli trials, let X be the random variable for the trial number that gave the first success.
- Then X is called a geometric random variable and its pmf is given as:

$$P(X = i) = p(1 - p)^{i-1}$$

Properties: Mode

- Let $X \sim \text{Binomial}(n, p)$.
- For $p = 0$, the mode is 0. For $p = 1$, the mode is n . Why?
- Consider the cases $0 < p < 1$ below.
- Then $P(X = k) \geq P(X = k-1) \leftrightarrow k \leq (n+1)p$ (prove this).
- Also $P(X = k) \geq P(X = k+1) \leftrightarrow k \geq (n+1)p-1$ (prove this).
- Any integer-valued k which satisfies both the above conditions will be the mode (why?).
- If $(n+1)p$ is an integer then $(n+1)p$ and $(n+1)p-1$ are both modes. Otherwise $\text{floor}((n+1)p)$ is the mode.

Poisson distribution

Definition – and genesis

- We have seen the binomial distribution before:

$$P(X = i) = \frac{n!}{i!(n-i)!} p^i (1-p)^{n-i}$$

- Here p is the success probability. We can express it in the form

$$p = \frac{\lambda}{n}, \lambda = \text{expected number of successes in } n \text{ trials}$$

- Hence

$$P(X = i) = \frac{n!}{i!(n-i)!} \left(\frac{\lambda}{n}\right)^i \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

Definition – and genesis

- We have

$$P(X = i) = \frac{n(n-1)(n-2)\dots(n-i+1)}{i!} \left(\frac{\lambda}{n}\right)^i \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

- In the limit when $n \rightarrow \infty$ and $p \rightarrow 0$ such that $\lambda = np$, we have

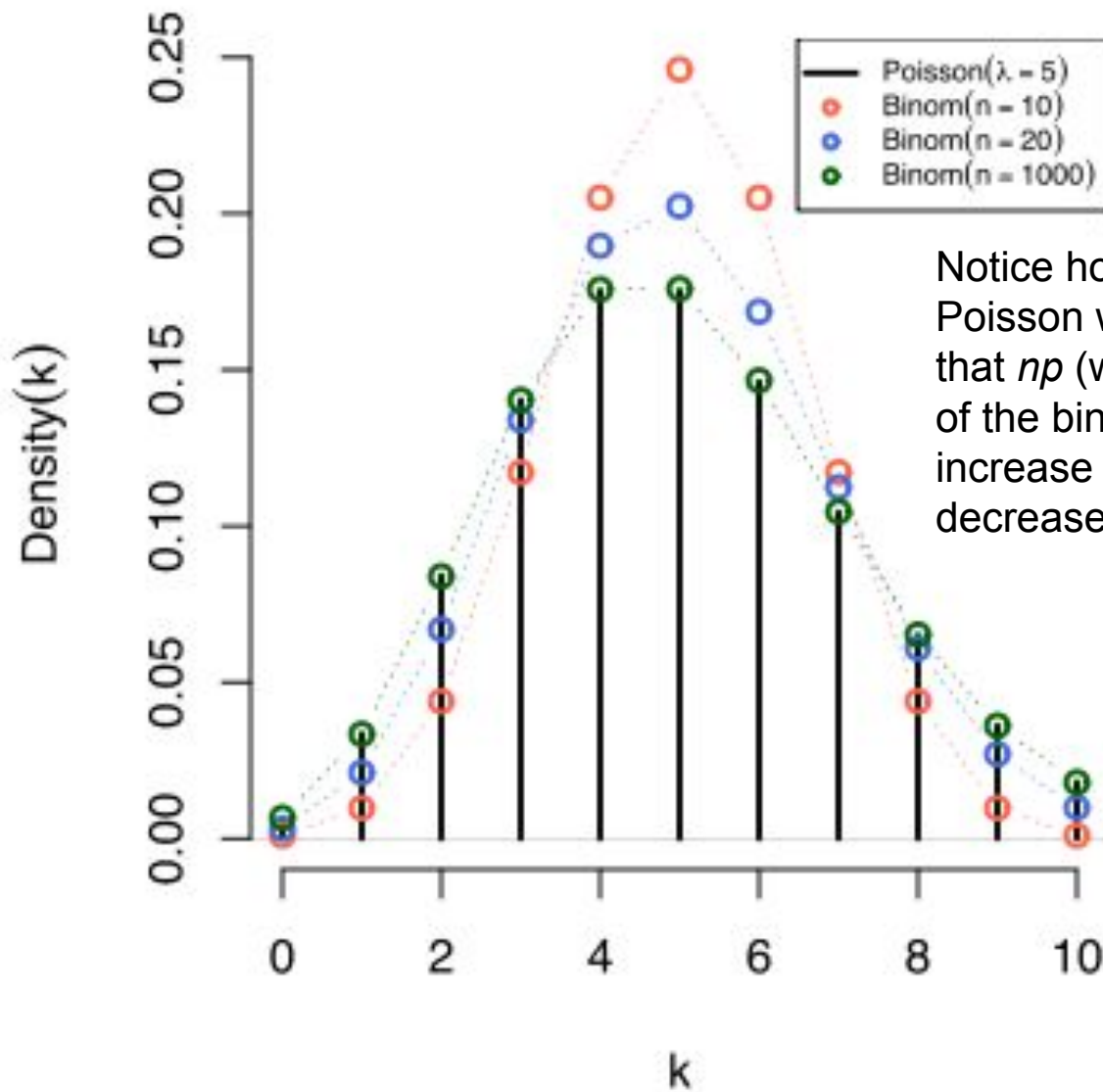
$$P(X = i) = \frac{\lambda^i}{i!} e^{-\lambda}$$

proof on the board

- This is called as the **Poisson pmf** and the above statement is called the **Poisson limit theorem**.

Definition

- The Poisson distribution is used to model the number of successes of a long sequence of independent Bernoulli trials if the *expected number of successes* (i.e. λ) is *known* and *constant*.
- For a Poisson random variable, note that the expected number of successes λ is constant and the **parameter** of the pmf.
- This is unlike the Binomial pmf for which the success probability p of a *single* Bernoulli trial is constant and also a parameter of the pmf.
- Note λ is strictly positive and real-valued, whereas the values of a Poisson random variable are non-negative integers.



Notice how the binomial is resembling the Poisson with an increase in n . Also notice that np (which is approximately the mode of the binomial) is constant despite increase in n since there is simultaneous decrease in p .

Properties

- To double check that it is indeed a valid pmf, we check that:

$$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} = e^{-\lambda} e^{\lambda} = 1$$

Using Taylor's series for exponential function about 0

- The afore-mentioned analysis tells us that the expected number of successes is equal to λ . To prove this rigorously – see next slide.

Properties

- The afore-mentioned analysis tells us that the expected number of successes is equal to λ . To prove this rigorously:

$$E(X) = \left(\sum_{i=0}^{\infty} i e^{-\lambda} \frac{\lambda^i}{i!} \right)$$

$$= \left(\sum_{i=1}^{\infty} i e^{-\lambda} \frac{\lambda^i}{i!} \right)$$

$$= \lambda e^{-\lambda} \left(\sum_{i=1}^{\infty} \frac{\lambda^{i-1}}{(i-1)!} \right)$$

$$= \lambda e^{-\lambda} \left(\sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \right)$$

$$= \lambda e^{-\lambda} e^{\lambda} = \lambda$$

Properties

- Variance:

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = \left(\sum_{i=0}^{\infty} i^2 e^{-\lambda} \frac{\lambda^i}{i!} \right) = \lambda^2 + \lambda$$

Detailed proof on the board. Also see [here](#).

$$\therefore \text{Var}(X) = \lambda$$

- MGF:

$$\Phi_X(t) = E(e^{tX})$$

$$= \sum_{i=0}^{\infty} e^{ti} e^{-\lambda} \lambda^i / i! = e^{-\lambda} \sum_{i=0}^{\infty} (\lambda e^t)^i / i!$$

$$= e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}$$

Properties

- **Mode:**

k is a mode if

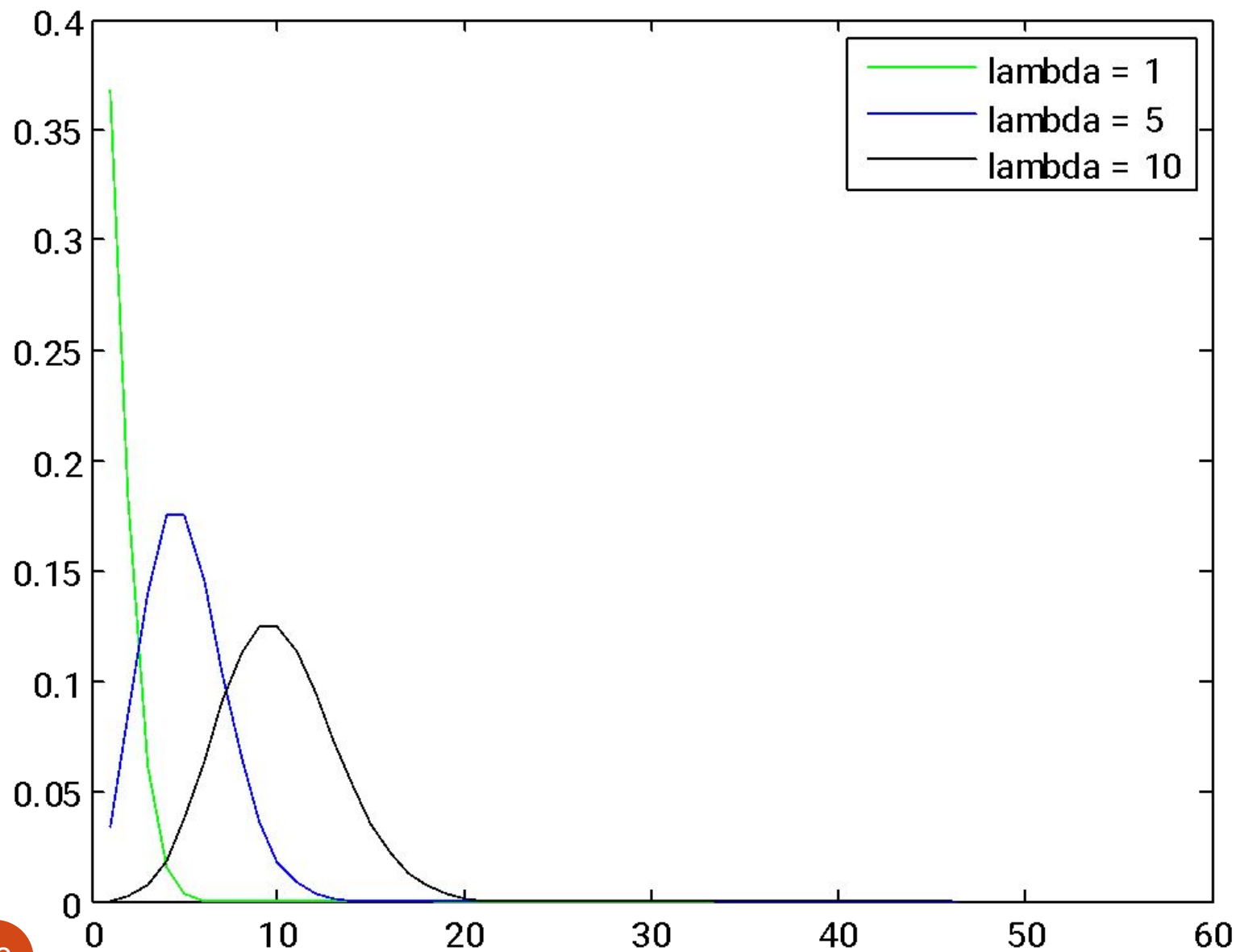
$$P(X = k) \geq P(X = k + 1) \rightarrow k \geq \lambda - 1$$

and

$$P(X = k) \geq P(X = k - 1) \rightarrow k \leq \lambda$$

Thus we seek an **integer** k that satisfies both these conditions - note that often, λ is **not** an integer.

Notice: the mean and variance both increase with λ .



Poisson approximated by a Gaussian

- From the previous plot, we can see that the Poisson distribution starts looking like the bell-curve.
- For large λ , $\text{Poisson}(\lambda)$ is approximated by $N(\lambda, \lambda)$.
- This can be proved by showing that the MGF of $(X - \lambda)/\sqrt{\lambda}$ limits to $\exp(t^2/2)$, i.e. the MGF of $N(0,1)$ as $\lambda \rightarrow \infty$.

Properties

- Consider independent Poisson random variables X and Y having parameters λ_1 and λ_2 respectively. Then $Z = X + Y$ is also a Poisson random variable with parameter $\lambda_1 + \lambda_2$.

Detailed proof on the board and in tutorial 1.

- PMF – recurrence relation:
$$P(X = i) = \frac{\lambda^i}{i!} e^{-\lambda}$$
$$P(X = i + 1) = \frac{\lambda^{i+1}}{(i + 1)!} e^{-\lambda}$$
$$\frac{P(X = i + 1)}{P(X = i)} = \frac{\lambda}{i + 1}, P(X = 0) = e^{-\lambda}$$

Properties

- If $X \sim \text{Poisson}(\lambda)$ and if $P(Y|X=l) = \text{Binomial}(l, p)$ where $\lambda > 0$ and $0 \leq p \leq 1$, then $Y \sim \text{Poisson}(\lambda p)$. This is called “**thinning of a Poisson random variable by a Binomial**”. *We will cover this derivation in a tutorial.*
- **Practical use:** Points in a scene being imaged send out photons of light at the rate of λ (Poisson). Of these, only a small fraction p manage to enter the camera (binomial). Hence the effective rate captured by the camera is λp .

Poisson distribution: examples

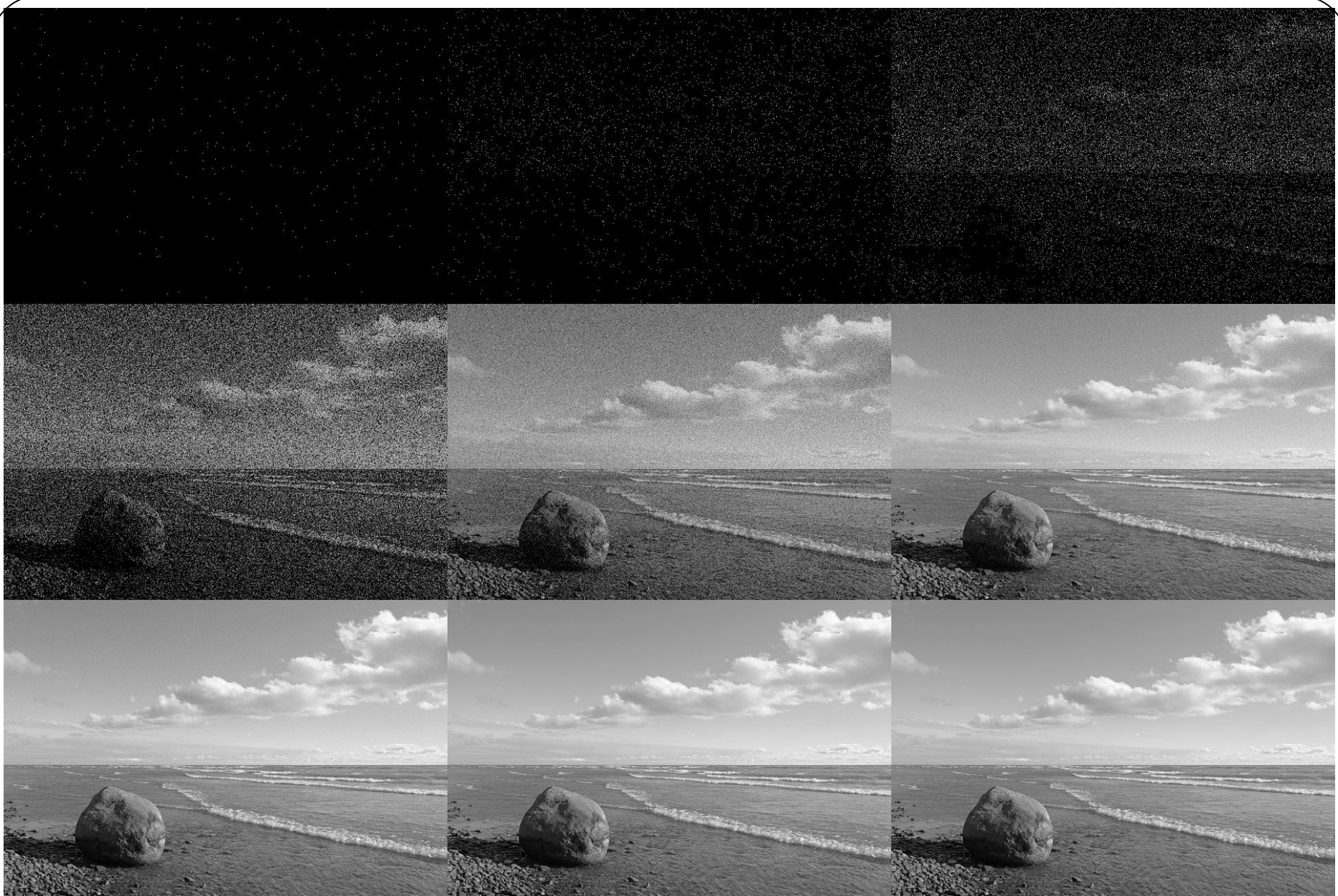
- Photon counting in a camera:
 - During the exposure time of the camera, the shutter is open
 - Photons of light enter the camera and hit the CMOS sensor array
 - Let us divide the exposure time into n time intervals
 - The probability of a photon hitting the sensor array in this time is λ/n .
 - Expected number of photons counted by the pixel = $n \times \lambda/n = \lambda$.
 - Why is λ a constant?
 - It is directly proportional to scene brightness, pixel area, exposure time and sensor efficiency – all of which are constant in a small time interval.
- The number of misprints in a book
 - Assume n words in the book
 - Assume each word has a small probability λ/n of being mis-typed
 - Expected number of mis-typed words = $n \times \lambda/n = \lambda$.
- In general, the Poisson distribution is used to model rare events, even though the event has plenty of “opportunities” to occur. (Sometimes called the **law of rare events** or the **law of small numbers**).

Poisson distribution: examples

- Number of people in a country who live up to 100 years
- Number of wrong numbers dialed in a day
- Number of laptops that fail on the first day of use

Poisson Noise

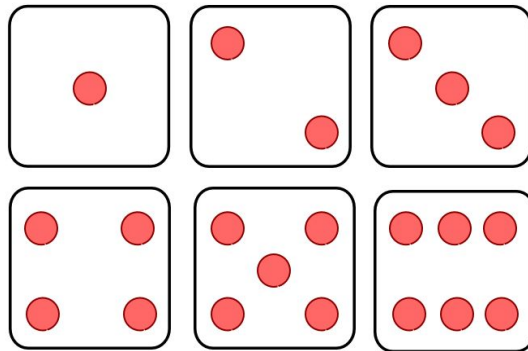
- The Poisson distribution models noise in optical image acquisition very well.
- Each camera detector acts as a photon counter.
- As ambient lighting increases, the variance of the Poisson noise increases (along with the mean), but the image becomes less noisy (why?)
- See next slide for visual examples.



Multinomial distribution

Definition

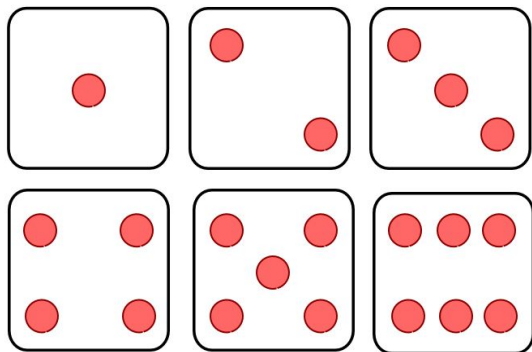
- Consider a sequence of n independent trials each of which will produce one out of k possible outcomes, where the set of possible outcomes is the *same* for each trial.



- Assume that the probability of each of the k outcomes is known and constant and given by $p_1, p_2, \dots, p_k$.

Definition

- Let $\mathbf{X}$ be a k -dimensional random variable for which the i^{th} element (X_i) represents the number of trials that produced the i^{th} outcome (also known as the number of *successes* for the i^{th} category)



Eg: in 20 throws of a die, you had 2 ones, 4 twos, 7 threes, 4 fours, 1 five and 2 sixes.

Definition

- Then the pmf of $\mathbf{X}$ is given as follows:

$$\begin{aligned} P(\mathbf{X} = (x_1, x_2, \dots, x_k)) \\ &= P(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k) \\ &= \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k} \end{aligned}$$

$$\forall i, 0 \leq p_i \leq 1, \sum_{i=1}^k p_i = 1, x_1 + x_2 + \dots + x_k = n$$

The number of combinations of n objects which can be divided into k groups of identical objects. There are x_1 objects of type 1, x_2 objects of type 2, and x_k objects of type k .

- This is called the **multinomial pmf**.

Definition

- The number of trials n and the success probabilities for each category, i.e. $p_1, p_2, \dots, p_k$ are all *parameters* of the multinomial pmf.
- If you know all success probabilities except one (say p_k), that is also no problem, as all probabilities sum up to 1.
- Hence some books consider only $n, p_1, p_2, \dots, p_{k-1}$ as parameters of the multinomial PMF.
- Remember: The multinomial random variable is a vector whose i^{th} component is the number of successes of the i^{th} category (i.e. the number of times that the trials produced a result of the i^{th} category).

Properties

- Mean **vector**:

$$E(\mathbf{X}) = (np_1, np_2, \dots, np_k), E(X_i) = np_i$$

- Variance of a component

$$Var(X_i) = Var\left(\sum_{j=1}^n X_{ij}\right) = \sum_{j=1}^n Var(X_{ij}) = np_i(1 - p_i)$$

Assuming independent trials

X_{ij} is a Bernoulli random variable which tells you whether or not there was a success in the i^{th} category on the j^{th} trial

Properties

- For vector-valued random variables, the variance is replaced by the **covariance matrix**. The covariance matrix **C** in this case will have size $k \times k$, where we have:

$$C(i, j) = E[(X_i - \mu_i)(X_j - \mu_j)] = \text{Cov}(X_i, X_j)$$

$$\text{Cov}(X_i, X_j) = -np_i p_j, i \neq j$$

$$= np_i(1 - p_i), i = j$$

Proof : next page

$$\text{Cov}(X_i, X_j) = -np_i p_j$$

Pr oof :

$$X_i = \# \text{ successes in category } i = \sum_{m=1}^n X_{im}$$

$$X_j = \# \text{ successes in category } j = \sum_{l=1}^n X_{jl}$$

These are independent Bernoulli random variables – each representing the outcome of a trial (indexed by m and l)

$$\text{Cov}(X_i, X_j) = \sum_{m=1}^n \sum_{l=1}^n \text{Cov}(X_{im}, X_{jl})$$

By linearity of covariance

$$= \sum_{m=1}^n \sum_{l=1, l \neq m}^n \text{Cov}(X_{im}, X_{jl}) + \sum_{l=1}^n \text{Cov}(X_{il}, X_{jl})$$

By independence of trials

$$= 0 + \sum_{l=1}^n (E(X_{il} X_{jl}) - E(X_{il}) E(X_{jl}))$$

Since in a trial, success can be achieved only in one category

$$= \sum_{l=1}^n (0 - p_i p_j) = -np_i p_j$$

MGF for a Multinomial

- For $k = 2$, the multinomial reduces to the binomial.
- Let us derive the MGF for $k = 3$ (trinomial):

$$\mathbf{X} = (X_1, X_2), \mathbf{x} = (x_1, x_2)$$

$$P(\mathbf{X} = \mathbf{x}) = \frac{n!}{x_1! x_2! (n - x_1 - x_2)!} p_1^{x_1} p_2^{x_2} (1 - p_1 - p_2)^{n - x_1 - x_2}$$

$$\Phi_{\mathbf{X}}(\mathbf{t}) = \Phi_{\mathbf{X}}(t_1, t_2) = E(e^{t_1 X_1 + t_2 X_2})$$

$$= \sum_{x_1=0}^n \sum_{x_2=0}^{n-x_1} \frac{n!}{x_1! x_2! (n - x_1 - x_2)!} p_1^{x_1} p_2^{x_2} (1 - p_1 - p_2)^{n - x_1 - x_2} e^{t_1 x_1} e^{t_2 x_2}$$

$$= \sum_{x_1=0}^n \sum_{x_2=0}^{n-x_1} \frac{n!}{x_1! x_2! (n - x_1 - x_2)!} (p_1 e^{t_1})^{x_1} (p_2 e^{t_2})^{x_2} (1 - p_1 - p_2)^{n - x_1 - x_2}$$

$$= (p_1 e^{t_1} + p_2 e^{t_2} + 1 - p_1 - p_2)^n$$

This follows from the multinomial theorem.

MGF for a Multinomial

- Multinomial theorem:

$$(x_1 + x_2 + \dots + x_m)^n = \sum_{k_1+k_2+\dots+k_m=n} \frac{n!}{k_1!k_2!\dots k_m!} \prod_{i=1}^m x_i^{k_i}$$

- For arbitrary k :

$$\mathbf{X} = (X_1, X_2, \dots, X_{k-1}), \mathbf{x} = (x_1, x_2, \dots, x_{k-1})$$

$$\Phi_{\mathbf{X}}(\mathbf{t}) = (p_1 e^{t_1} + p_2 e^{t_2} + \dots + p_{k-1} e^{t_{k-1}} + 1 - p_1 - p_2 - \dots - p_{k-1})^n$$

- We are dropping X_k for the same reason as mentioned while discussing multinomial PMF parameters on previous slides. But you can surely include X_k as well, in which the MGF will be:

$$\mathbf{X} = (X_1, X_2, \dots, X_{k-1}, X_k), \mathbf{x} = (x_1, x_2, \dots, x_{k-1}, x_k)$$

$$\Phi_{\mathbf{X}}(\mathbf{t}) = (p_1 e^{t_1} + p_2 e^{t_2} + \dots + p_{k-1} e^{t_{k-1}} + p_k e^{t_k})^n$$

MGF for multinomial

- To derive the mean of the multinomials using MGFs:

$$\mathbf{X} = (X_1, X_2, \dots, X_{k-1}, X_k), \mathbf{x} = (x_1, x_2, \dots, x_{k-1}, x_k)$$

$$\Phi_X(\mathbf{t}) = (p_1 e^{t_1} + p_2 e^{t_2} + \dots + p_{k-1} e^{t_{k-1}} + p_k e^{t_k})^n$$

$$\frac{\partial \Phi_X(\mathbf{t})}{\partial t_i} = n(p_1 e^{t_1} + p_2 e^{t_2} + \dots + p_{k-1} e^{t_{k-1}} + p_k e^{t_k})^{n-1} p_i e^{t_i} \text{ where } 1 \leq i \leq k$$

$$\left. \frac{\partial \Phi_X(\mathbf{t})}{\partial t_i} \right|_{\mathbf{t}=\mathbf{0}} = np_i$$

- Try the MGF method to derive the covariance matrix of a multinomial distribution.

Hypergeometric distribution

Sampling with and without replacement

- Suppose there are k objects each of a different type.
- When you sample 2 objects from these **with replacement**, you pick a particular object with probability $1/k$, and you place it back (**replace** it).
- The probability of picking an object of another type is again $1/k$.
- When you sample **without replacement**, the probability that your first object was of so and so type is $1/k$. The probability that your second object was of so and so type is now $1/(k-1)$ because you **didn't** put the first object back!

Definition

- Consider a set of objects of which N are of good quality and M are defective.
- Suppose you pick some n objects out of these *without* replacement.
- There are $C(N+M, n)$ ways of doing this.
- Let X be a random variable denoting the number of good quality objects picked (out of a total of n).

Definition

- There are $C(N,i)C(M,n-i)$ ways to pick i good quality objects and $n-i$ bad objects.
- So we have

$$P(X = i) = \frac{C(N,i)C(M,n-i)}{C(N+M,n)}, 0 \leq i \leq n$$

$$C(a,b) = 0 \text{ if } b > a \text{ or } b < 0$$

- Such a random variable X is called a **hypergeometric random variable**.

Properties

- Consider random variable X_i which has value 1 if the i^{th} trial produces a good quality object and 0 otherwise.
- Now consider the following probabilities:

$$P(X_1 = 1) = \frac{N}{N + M}$$

$$P(X_2 = 1) = P(X_2 = 1 | X_1 = 1)P(X_1 = 1) +$$

$$P(X_2 = 1 | X_1 = 0)P(X_1 = 0)$$

$$= \frac{N-1}{N+M-1} \frac{N}{N+M} + \frac{N}{N+M-1} \frac{M}{N+M} = \frac{N}{N+M}$$

$$\text{In general, } P(X_i = 1) = \frac{N}{N+M}$$

Properties

- Note that:

$$X = \sum_{i=1}^n X_i$$

Each X_i is a Bernoulli random variable with parameter $p=N/(N+M)$.

$$\therefore E(X) = E\left(\sum_{i=1}^n X_i\right) = \frac{nN}{N+M}$$

$$\therefore Var(X) = Var\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n Var(X_i) + 2 \sum_{i=1}^n \sum_{j=i+1}^n Cov(X_i, X_j)$$

$$Var(X_i) = P(X_i = 1)(1 - P(X_i = 1)) = \frac{NM}{N+M}$$

Properties

- Note that:

$$\therefore \text{Var}(X) = \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i=1}^n \sum_{j=i+1}^n \text{Cov}(X_i, X_j)$$

$$\text{Var}(X_i) = P(X_i = 1)(1 - P(X_i = 1)) = \frac{NM}{N + M}$$

$$\text{Cov}(X_i, X_j) = E(X_i X_j) - E(X_i)E(X_j)$$

$$E(X_i X_j) = P(X_i X_j = 1) = P(X_j = 1, X_i = 1)$$

$$= P(X_j = 1 | X_i = 1)P(X_i = 1)$$

$$= \frac{N-1}{N+M-1} \frac{N}{N+M}$$

Properties

- Note that:

$$\text{Cov}(X_i, X_j) = E(X_i X_j) - E(X_i)E(X_j)$$

$$E(X_i X_j) = P(X_i X_j = 1) = P(X_j = 1, X_i = 1)$$

$$= P(X_j = 1 | X_i = 1)P(X_i = 1)$$

$$= \frac{N-1}{N+M-1} \frac{N}{N+M}$$

$$\text{Cov}(X_i, X_j) = \frac{N(N-1)}{(N+M)(N+M-1)} - \left(\frac{N}{N+M} \right)^2$$

$$= \frac{-NM}{(N+M)^2(N+M-1)}$$

Properties

- Note that:

$$\begin{aligned} \text{Cov}(X_i, X_j) &= \frac{N(N-1)}{(N+M)(N+M-1)} - \left(\frac{N}{N+M} \right)^2 \\ &= \frac{-NM}{(N+M)^2(N+M-1)} \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= \frac{nNM}{(N+M)^2} - \frac{n(n-1)NM}{(N+M)^2(N+M-1)} \\ &= \frac{nNM}{(N+M)^2} \left(1 - \frac{n-1}{N+M-1} \right) \\ &= np(1-p) \left(1 - \frac{n-1}{N+M-1} \right) \end{aligned}$$

$\approx np(1-p)$ when N and/or M is/are very large

Recall: Each X_i is a Bernoulli random variable with parameter $p = N/(N+M)$.

Hypergeometric distribution in Ecology

- Consider the task of finding the number of leopards in SGNP (Sanjay Gandhi National Park).
- Let N = true number of leopards in SGNP.
- Ecologists use a method called capture-recapture.
- First some r leopards are captured, marked (with a *collar*) and then released.
- Then a new catch of size n is made, and the number of marked animals in this catch (X) is determined.
- Then we have:

$$P(X = i) = \frac{C(r, i)C(N - r, n - i)}{C(N, n)}$$

Assumptions:(1) N does not change between the time of the two catches, (2) each leopard is equally likely to be caught as the others.

Hypergeometric distribution in Ecology

- The fraction of marked animals in the new catch = i/n .
- The fraction of marked animals in SGNP = r/N .
- Then we have: $i/n = r/N$ which is used to calculate N
 $\rightarrow N = rn/i$.
- It turns out the estimate rn/i is approximately the one that makes N the most likely given $X = i$.
- Why? Because we want $P(X=i;N) > P(X=i;N-1)$ which gives $N < rn/i$.
- And we want $P(X=i;N) > P(X=i;N+1)$ which gives $N > nr/i - 1$.
- Finally $N = \text{floor}(nr/i)$.

Gaussian distribution

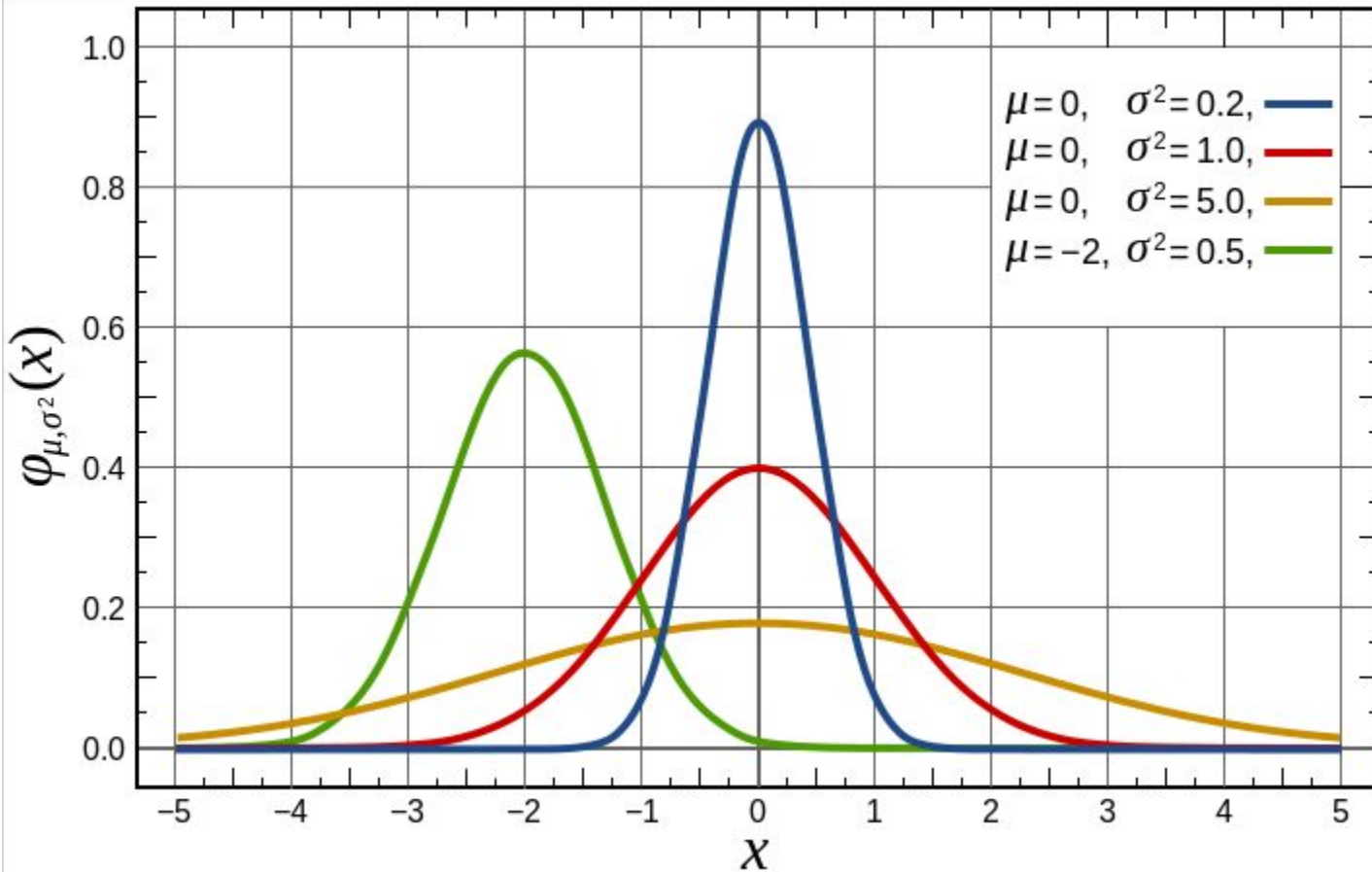
Definition

- A continuous random variable is said to be normally distributed with parameters mean μ and standard deviation σ if it has a probability density function given as:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \text{ denoted as}$$

$$N(\mu, \sigma^2)$$

- This pdf is symmetric about the mean μ and has the shape of the “bell curve”.



https://upload.wikimedia.org/wikipedia/commons/7/74/Normal_Distribution_PDF.svg

Definition

- If $\mu=0$ and $\sigma=1$, it is called the **standard normal distribution**

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \text{ denoted as } N(0,1)$$

- To verify that this is a valid pdf:

$$\left(\int_{-\infty}^{\infty} e^{-x^2} dx \right)^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta$$

Note the change from (x,y) to polar coordinates (r,θ) .
 $x = r \cos(\theta)$
 $y = r \sin(\theta)$

$$= 2\pi \int_0^{\infty} e^{-r^2} r dr = \pi \int_0^{\infty} e^{-s} ds = \pi$$

$$\therefore \frac{1}{(1/\sqrt{2})\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2} dx = 1$$

→ This is a Gaussian pdf with mean 0 and standard deviation $1/\sqrt{2}$. Thus we have verified that this particular Gaussian function is a valid pdf. You can verify that Gaussians with arbitrary mean and variance are valid pdfs by a change of variables.

Properties

- Mean:

$$E(X - \mu) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x - \mu) e^{-(x-\mu)^2/(2\sigma^2)} dx$$

$$= \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (y) e^{-(y)^2/(2)} dy$$

$$= 0 - - - why?$$

$$\therefore E[X] = \mu$$

Properties

- Variance:

$$E((X - \mu)^2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x - \mu)^2 e^{-(x-\mu)^2/(2\sigma^2)} dx$$

$$= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^2 e^{-(y)^2/(2)} dy = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\underbrace{y}_u) (\underbrace{ye^{-(y)^2/(2)}}_{v du}) dy$$

$$= \frac{\sigma^2}{\sqrt{2\pi}} \left[\left(-ye^{-y^2/2} \right)_{-\infty}^{\infty} - \int_{-\infty}^{\infty} -e^{-y^2/2} dy \right]$$

$$= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-y^2/2} dy = \frac{\sigma^2}{\sqrt{2\pi}} \sqrt{2\pi} = \sigma^2$$

$$\int u dv = uv - \int v du$$

$$\int ye^{-y^2/2} dy = -e^{-y^2/2}$$

Properties

If $X \sim N(\mu, \sigma^2)$ and if $Y = aX + b$, then

$$Y \sim N(a\mu + b, a^2\sigma^2)$$

Proof on board. And in the book.

Properties

- Median = mean (why?)
- Because of symmetry of the pdf about the mean
- Mode = mean – can be checked by setting the first derivative of the pdf to 0 and solving, and checking the sign of the second derivative.
- CDF for a 0 mean Gaussian with variance 1 – is given by:

$$\Phi(x) = F_X(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-z^2/2} dz$$

Properties

- CDF – it is given by:

$$\Phi(x) = F_X(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-(z)^2/(2)} dz$$

- It is closely related to the error function $\text{erf}(x)$ defined as:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-z^2} dz$$

$$\Phi(x) = \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \int_0^x e^{-z^2/2} dz$$

- It follows that:

$$\Phi(x) = \frac{1}{2} \left[1 + \text{erf}\left(\frac{x}{\sqrt{2}}\right) \right]$$

$$= \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \int_0^{x/\sqrt{2}} e^{-t^2} dt \sqrt{2}$$

$$= \frac{1}{2} + \frac{1}{2} \text{erf}\left(\frac{x}{\sqrt{2}}\right)$$

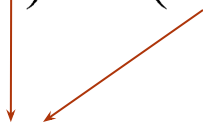
Properties

- For a Gaussian with mean μ and standard deviation σ , it follows that:

$$\Phi\left(\frac{x - \mu}{\sigma}\right) = \frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{x - \mu}{\sigma\sqrt{2}}\right) \right]$$

- The probability that a Gaussian random variable has values from $\mu - n\sigma$ to $\mu + n\sigma$ is given by:

$$\Phi(n) - \Phi(-n) = \frac{1}{2} \operatorname{erf}\left(\frac{n}{\sqrt{2}}\right) - \frac{1}{2} \operatorname{erf}\left(\frac{-n}{\sqrt{2}}\right) = \operatorname{erf}\left(\frac{n}{\sqrt{2}}\right)$$

 If $X \sim N(\mu, \sigma^2)$, then these are the values of $(X - \mu) / \sigma$.

Properties

- The probability that a Gaussian random variable has values from $\mu - n\sigma$ to $\mu + n\sigma$ is given by:

$$\Phi(n) - \Phi(-n) = \operatorname{erf}\left(\frac{n}{\sqrt{2}}\right)$$

n	$\Phi(n) - \Phi(-n)$
1	68.2%
2	95.4%
3	99.7%
4	99.99366%
5	99.9999%
6	99.9999998%

Hence a Gaussian random variable lies within $\pm 3\sigma$ from its mean with more than 99% probability

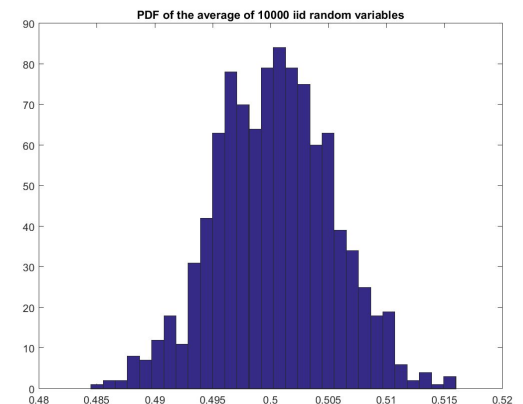
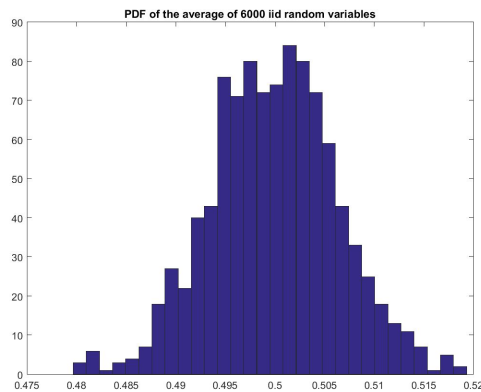
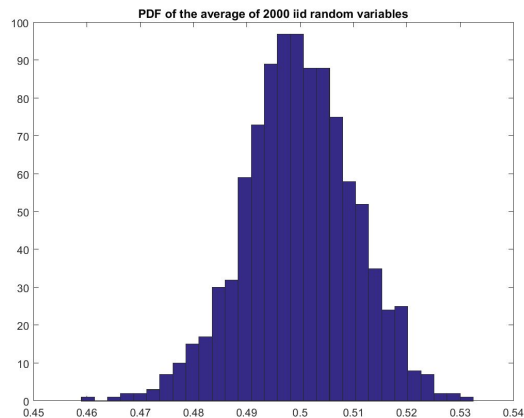
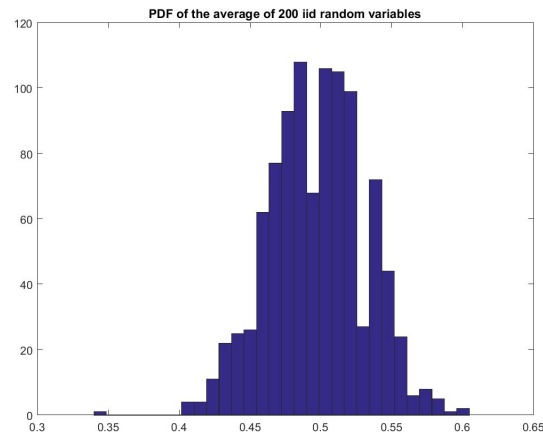
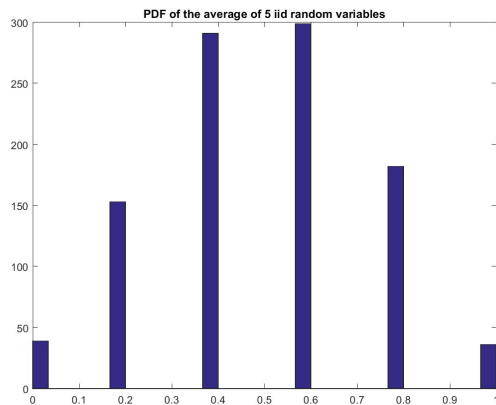
Properties

- MGF:

$$\Phi_X(t) = \exp(\mu t + \sigma^2 t^2 / 2) \quad \underline{\text{Proof here.}}$$

A strange phenomenon

- Suppose I toss a coin some n times independently and count the number of heads and divide it by n – call it x_1 .
- We will term this set of n independent coin tosses as one ROUND.
- I repeat this experiment for several (say T) rounds.
- In each round, I count the number of heads in n independent coin tosses and divide it by n
- This produces the value x_t for the t^{th} round ($t = 1$ to T).
- I now plots a histogram given the data $\{x_1, x_2, \dots, x_T\}$.
- What does the histogram look like?



With $n = 5, 200, 2000, 6000, 10000$ (Left to right, top to bottom)
 Note: these plots are for the number of heads DIVIDED by n

A strange phenomenon

- Let's say you draw $n = 2$ values, called x_1 and x_2 , from a $[0,1]$ uniform random distribution and compute:

$$y_j = \frac{\sum_{i=1}^n x_i}{n}$$

Sampling index = $i, 1 \leq i \leq n$
Iteration index = $j, 1 \leq j \leq m$

- You repeat this process some $m=5000$ times (say), and then plot the histogram of the computed $\{y_j, 1 \leq j \leq m\}$, values.
- Now suppose you repeat the earlier two steps with larger and larger n .

A strange phenomenon

- Now suppose you repeat the earlier two steps with larger and larger n .
- It turns out that as n grows larger and larger, the histogram starts resembling a 0 mean Gaussian distribution with variance equal to that of the sampling distribution (i.e. the $[0,1]$ distribution) divided by n .
- Now if you repeat the experiment with samples drawn from any other distribution instead of $[0,1]$ uniform random (i.e. you change the sampling distribution).
- The phenomenon still occurs, though the resemblance may start showing up at smaller or larger values of n .
- This leads us to a very interesting theorem called the **central limit theorem**.
- [Demo code link.](#)

Central limit theorem

- Consider $X_1, X_2, \dots, X_n$ to be a sequence of **independent** and **identically distributed** (i.i.d.) random variables each with mean μ and variance $\sigma^2 < \infty$. Then as $n \rightarrow \infty$, the **distribution** (i.e. CDF) of the following quantity:

$$Y_n = \sqrt{n} \left(\frac{\sum_{i=1}^n X_i}{n} - \mu \right)$$

converges to that of $N(0, \sigma^2)$. Or, we say Y_n converges in distribution to $N(0, \sigma^2)$. This is called the **Lindeberg-Levy central limit theorem**.

Central limit theorem: some comments

- Note that the random variables $X_1, X_2, \dots, X_n$ must be independent and identically distributed.
- Converge in distribution to a Gaussian means the following:

$$\lim_{n \rightarrow \infty} P(Y_n \leq z) = \Phi(z / \sigma)$$

- There is a version of the central limit theorem that requires **only independence** – and allows the random variables to belong to **different distributions**. This extension is called the **Lindeberg Central Limit theorem**, and is given on the next slide.

Lindeberg's Central limit theorem

- Consider $X_1, X_2, \dots, X_n$ to be a sequence of independent random variables each with mean μ_i and variance $(\sigma_i)^2 < \infty$. Then as $n \rightarrow \infty$, the distribution of the following quantity:

$$Y_n = \frac{\sum_{i=1}^n (X_i - \mu_i)}{\sqrt{\sum_{i=1}^n \sigma_i^2}}$$

$$\mathbf{1}_A : X \rightarrow \{0,1\}$$

$$\mathbf{1}_A(x) = 1 \text{ if } x \in A$$

$$= 0 \text{ otherwise}$$

Indicator
function

converges to that of $N(0, 1)$, provided for every $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} \left(\frac{\sum_{i=1}^n E[(X_i - \mu_i)^2 \cdot \mathbf{1}_{\{|X_i - \mu_i| > \varepsilon s_n\}}(X_i)]}{s_n} \right) = 0, s_n = \sum_{i=1}^n \sigma_i^2$$

Lindeberg's Central limit theorem

- Informally speaking, the take home message from the previous slide is that the CLT is valid even if the random variables emerge from different distributions.
- This provides a major motivation for the widespread usage of the Gaussian distribution.
- The errors in experimental observations are often modelled as Gaussian – because these errors often stem from many different independent sources, and are modelled as being weighted combinations of errors from each such source.
- The condition (for the indicator function) in the previous slide is that the variance of no one random variable, or groups of random variables, is allowed to dominate the sum s_n .

A few historical remarks (source Wikipedia)

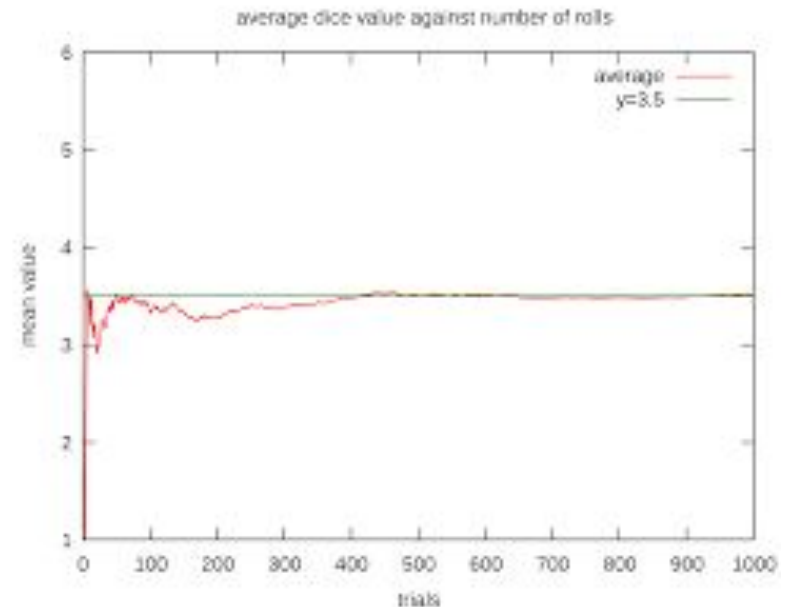
- An opinion about the CLT (due to Sir Francis Galton):

I know of **scarcely anything so apt to impress the imagination as the wonderful form of cosmic order** expressed by the "Law of Frequency of Error". The law would have been personified by the Greeks and deified, if they had known of it. It reigns with serenity and in complete self-effacement, amidst the wildest confusion. **The huger the mob, and the greater the apparent anarchy, the more perfect is its sway. It is the supreme law of Unreason.** Whenever a large sample of chaotic elements are taken in hand and marshalled in the order of their magnitude, an unsuspected and most beautiful form of regularity proves to have been latent all along.

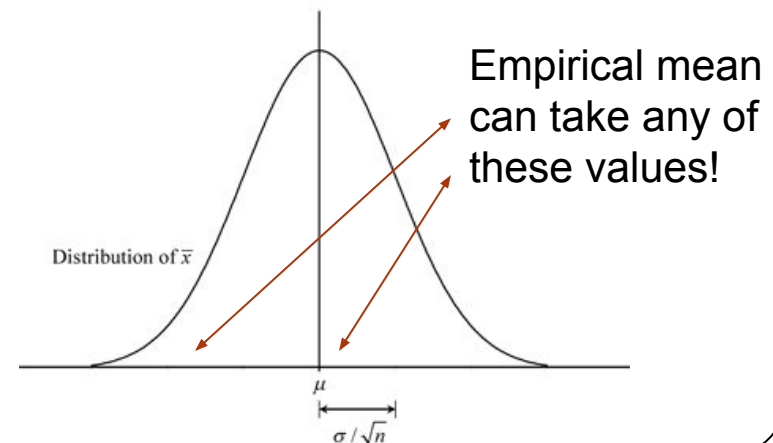
- The CLT was proved first in 1922. Alan Turing independently proved it for his bachelor's thesis in 1934 at Cambridge University (King's college), unaware of its publication before.

Central limit theorem versus law of large numbers

- The law of large numbers says that the empirical mean calculated from a large number of samples is **equal to** (or very close) to the *true mean* μ (of the distribution from which the samples were drawn).
- The central limit theorem says that the empirical mean calculated from a large number of samples is a **random variable** following a Gaussian distribution with mean equal to the true mean μ (of the distribution from which the samples were drawn).

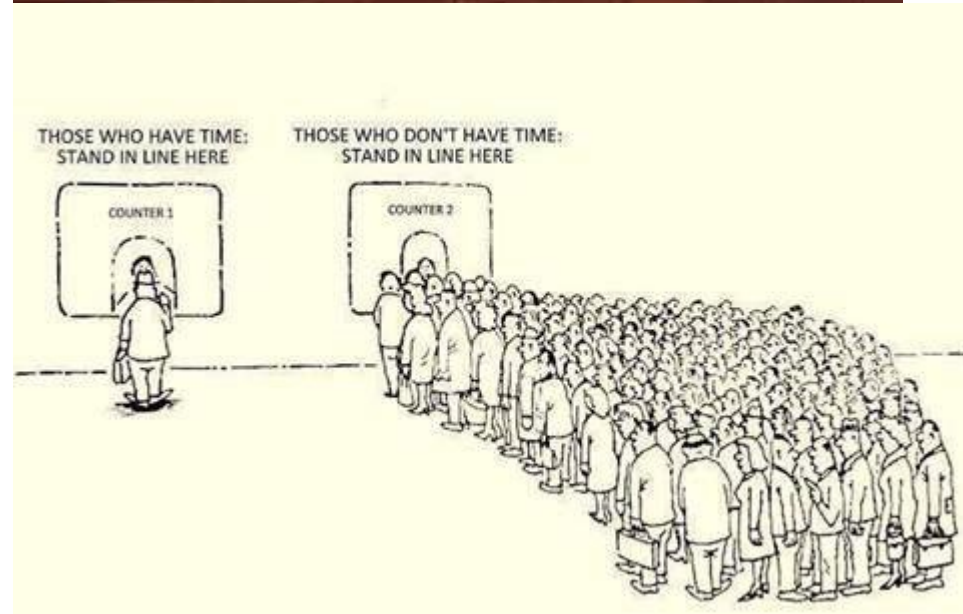


source



Central limit theorem versus law of large numbers

- Is this a contradiction?



Central limit theorem versus law of large numbers

- The answer is NO!
- Go and look back at the central limit theorem.

$$Y_n = \sqrt{n} \left(\frac{\sum_{i=1}^n Y_i}{n} - \mu \right) \sim N(0, \sigma^2)$$

$$\rightarrow \left(\frac{\sum_{i=1}^n X_i}{n} - \mu \right) \sim N(0, \sigma^2 / n) \quad (--- why?)$$

$$\rightarrow \left(\frac{\sum_{i=1}^n X_i}{n} \right) \sim N(\mu, \sigma^2 / n)$$

This variance drops to 0 when n is very large! All the probability is now concentrated at the mean!

Proof of Lindberg-Levy CLT using MGFs

- Consider the n i.i.d. random variables $X_1, X_2, \dots, X_n$ with mean μ and variance σ^2 . Let their sum be S_n .
- Then we have to prove that:

$$\lim_{n \rightarrow \infty} P\left(\frac{S_n - \mu n}{\sigma \sqrt{n}} \leq x\right) = \int_{-\infty}^x \frac{e^{-y^2/2}}{\sqrt{2\pi}} dy$$

- For that we will prove that the MGF of Z_n equals the MGF of the standard normal distribution (i.e. $\exp(t^2/2)$) where

$$Z_n = \left(\frac{S_n - \mu n}{\sigma \sqrt{n}}\right)$$

Proof of Lindberg-Levy CLT using MGFs

If X_1, X_2 are independent, then

$$\phi_{X_1+X_2}(t) = \phi_{X_1}(t)\phi_{X_2}(t)$$

- By properties of the MGF, we have:

$$\phi_{S_n - \mu n}(t) = (\phi_{\tilde{X}}(t))^n \text{ where } \tilde{X} = (X - \mu)$$

$$\therefore \phi_{Z_n}(t) = \left(\phi_{\tilde{X}}\left(\frac{t}{\sigma\sqrt{n}}\right) \right)^n$$

Recall : If $Y = aX + b$,

$$\phi_Y(t) = e^{tb} \phi_X(at)$$

$\tilde{X}$ -tilde is used to denote any one of the iid RVs $X_1 - \mu, X_2 - \mu, \dots, X_n - \mu$

- We need to prove that:

$$\lim_{n \rightarrow \infty} n \log \left(\phi_{\tilde{X}}\left(\frac{t}{\sigma\sqrt{n}}\right) \right) = t^2 / 2$$

Recall : for a Gaussian r.v. X with mean μ and std. dev. σ ,

$$\Phi_X(t) = \exp(\mu t + \sigma^2 t^2 / 2)$$

Proof of Lindberg-Levy CLT using MGFs

L'Hospital's rule

Change of variables:
 $x = 1/\sqrt{n}$

$$\lim_{x \rightarrow 0} \frac{\log \phi_{\tilde{X}}(tx/\sigma)}{x^2} = \lim_{x \rightarrow 0} \frac{\left(\frac{\frac{d}{dx}(\phi_{\tilde{X}}(tx/\sigma))}{\phi_{\tilde{X}}(tx/\sigma)} \right)}{2x} = \lim_{x \rightarrow 0} \frac{E(e^{tx\tilde{X}/\sigma} t\tilde{X}/\sigma)}{2x\phi_{\tilde{X}}(tx/\sigma)}$$

Proof of Lindberg-Levy CLT using MGFs

L'Hospital's rule

Change of variables:
 $x = 1/\sqrt{n}$

$$\lim_{x \rightarrow 0} \frac{\log \phi_{\tilde{X}}(tx/\sigma)}{x^2} = \lim_{x \rightarrow 0} \frac{\left(\frac{\frac{d}{dx}(\phi_{\tilde{X}}(tx/\sigma))}{\phi_{\tilde{X}}(tx/\sigma)} \right)}{2x} = \lim_{x \rightarrow 0} \frac{E(e^{tx\tilde{X}/\sigma} t\tilde{X}/\sigma)}{2x\phi_{\tilde{X}}(tx/\sigma)}$$

(Note: $\frac{d}{dx}(E(e^{tx\tilde{X}/\sigma})) = E(e^{tx\tilde{X}/\sigma} t\tilde{X}/\sigma)$ by exchanging order of expectation and differentiation.

Putting $x = 0$, the numerator $= E(t\tilde{X}/\sigma) = tE(\tilde{X})/\sigma = tE(X - \mu)/\sigma = 0$.

The denominator is also 0 at $x = 0$.

This satisfies the condition to apply L'Hospital's rule)

Proof of Lindberg-Levy CLT using MGFs

$$LHS = \lim_{x \rightarrow 0} \frac{\log \phi_{\tilde{X}}(tx / \sigma)}{x^2} = \lim_{x \rightarrow 0} \frac{E\left(e^{tx\tilde{X}/\sigma} t\tilde{X} / \sigma\right)}{2x\phi_{\tilde{X}}(tx / \sigma)}$$

Continuing the proof from previous steps.

$$LHS = \lim_{x \rightarrow 0} \frac{\frac{d}{dx} \left(E\left(e^{tx\tilde{X}/\sigma} t\tilde{X} / \sigma\right) \right)}{2\phi_{\tilde{X}}(tx / \sigma) + 2x \frac{d}{dx} \left(\phi_{\tilde{X}}(tx / \sigma) \right)}$$

We will now apply L'Hospital's rule again. Note that the denominator is 0 at $x = 0$. The numerator is also 0 when $x = 0$ since $E(X\text{-tilde}) = 0$.

Proof of Lindberg-Levy CLT using MGFs

$$LHS = \lim_{x \rightarrow 0} \frac{\log \phi_{\tilde{X}}(tx / \sigma)}{x^2} = \lim_{x \rightarrow 0} \frac{E\left(e^{tx\tilde{X}/\sigma} t\tilde{X} / \sigma\right)}{2x\phi_{\tilde{X}}(tx / \sigma)}$$

We will now apply L'Hospital's rule again. Note that the denominator is 0 at $x = 0$. The numerator is also 0 when $x = 0$ since $E(X\text{-tilde}) = 0$.

Continuing the proof from previous steps.

$$\begin{aligned} LHS &= \lim_{x \rightarrow 0} \frac{\frac{d}{dx} \left(E\left(e^{tx\tilde{X}/\sigma} t\tilde{X} / \sigma\right) \right)}{2\phi_{\tilde{X}}(tx / \sigma) + 2x \frac{d}{dx} (\phi_{\tilde{X}}(tx / \sigma))} \\ &= \lim_{x \rightarrow 0} \frac{E\left(e^{tx\tilde{X}/\sigma} (t\tilde{X} / \sigma)^2\right)}{2\phi_{\tilde{X}}(tx / \sigma) + 2x \frac{d}{dx} (\phi_{\tilde{X}}(tx / \sigma))} = \frac{E\left((t\tilde{X} / \sigma)^2\right)}{2\phi_{\tilde{X}}(0)} \end{aligned}$$

This is 1.

Proof of Lindberg-Levy CLT using MGFs

$$LHS = \lim_{x \rightarrow 0} \frac{\log \phi_{\tilde{X}}(tx / \sigma)}{x^2} = \lim_{x \rightarrow 0} \frac{E\left(e^{tx\tilde{X}/\sigma} t\tilde{X} / \sigma\right)}{2x\phi_{\tilde{X}}(tx / \sigma)}$$

We will now apply L'Hospital's rule again. Note that the denominator is 0 at $x = 0$. The numerator is also 0 when $x = 0$ since $E(\tilde{X}) = 0$.

Continuing the proof from previous steps.

$$\begin{aligned} LHS &= \lim_{x \rightarrow 0} \frac{\frac{d}{dx} \left(E\left(e^{tx\tilde{X}/\sigma} t\tilde{X} / \sigma\right) \right)}{2\phi_{\tilde{X}}(tx / \sigma) + 2x \frac{d}{dx} (\phi_{\tilde{X}}(tx / \sigma))} \\ &= \lim_{x \rightarrow 0} \frac{E\left(e^{tx\tilde{X}/\sigma} (t\tilde{X} / \sigma)^2\right)}{2\phi_{\tilde{X}}(tx / \sigma) + 2x \frac{d}{dx} (\phi_{\tilde{X}}(tx / \sigma))} = \frac{E\left((t\tilde{X} / \sigma)^2\right)}{2\phi_{\tilde{X}}(0)} \end{aligned}$$

This is 1.

$$= \frac{t^2}{2\sigma^2} E(\tilde{X}^2) = \frac{t^2}{2\sigma^2} \sigma^2 = \frac{t^2}{2}$$

$\tilde{X}$ has 0 mean, so second moment equals variance

Application

- Your friend tells you that in 10000 successive independent unbiased coin tosses, he counted at least 5200 heads. Is (s)he serious or joking?
- **Answer:** Let $X_1, X_2, \dots, X_n$ be the random variables indicating whether or not the coin toss was a success (a heads).
- These are i.i.d. random variables whose sum is a random variable with:
 - mean $n\mu = 10000(0.5) = 5000$
 - standard deviation $\sigma n^{1/2} = \text{sqrt}(0.5(1-0.5))\text{sqrt}(10000) = 50$ (both by property of the binomial distribution).

Application

- Your friend tells you that in 10000 successive independent unbiased coin tosses, he counted at least 5200 heads. Is (s)he serious or joking?
- **Answer:** The given number of heads is at least 5200 which is 4 standard deviations away from the mean.
- The chance of 5200+ heads is less than 0.00001 (see the slide on error functions) since the total number of heads is a Gaussian random variable (as per central limit theorem).
- So your friend is (most likely) joking.
- Notice that this answer is much more principled than giving an answer purely based on some arbitrary threshold over $|X-5000|$.
- You will study much more of this when you do a topic called hypothesis testing.
- This result via the CLT is stronger than what you'd have got via the Chebyshev's inequality (how and why?)

Gaussian Tail Bounds

- It is interesting to derive bounds on the value of $P(X > x) = 1 - F_X(x)$ for Gaussian r.v. X .
- This is called the tail probability of X . It is given as follows:

$$\text{For } x > 0, P(X > x) = \int_x^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \leq \int_x^{\infty} \frac{t}{x\sqrt{2\pi}} e^{-t^2/2} dt$$

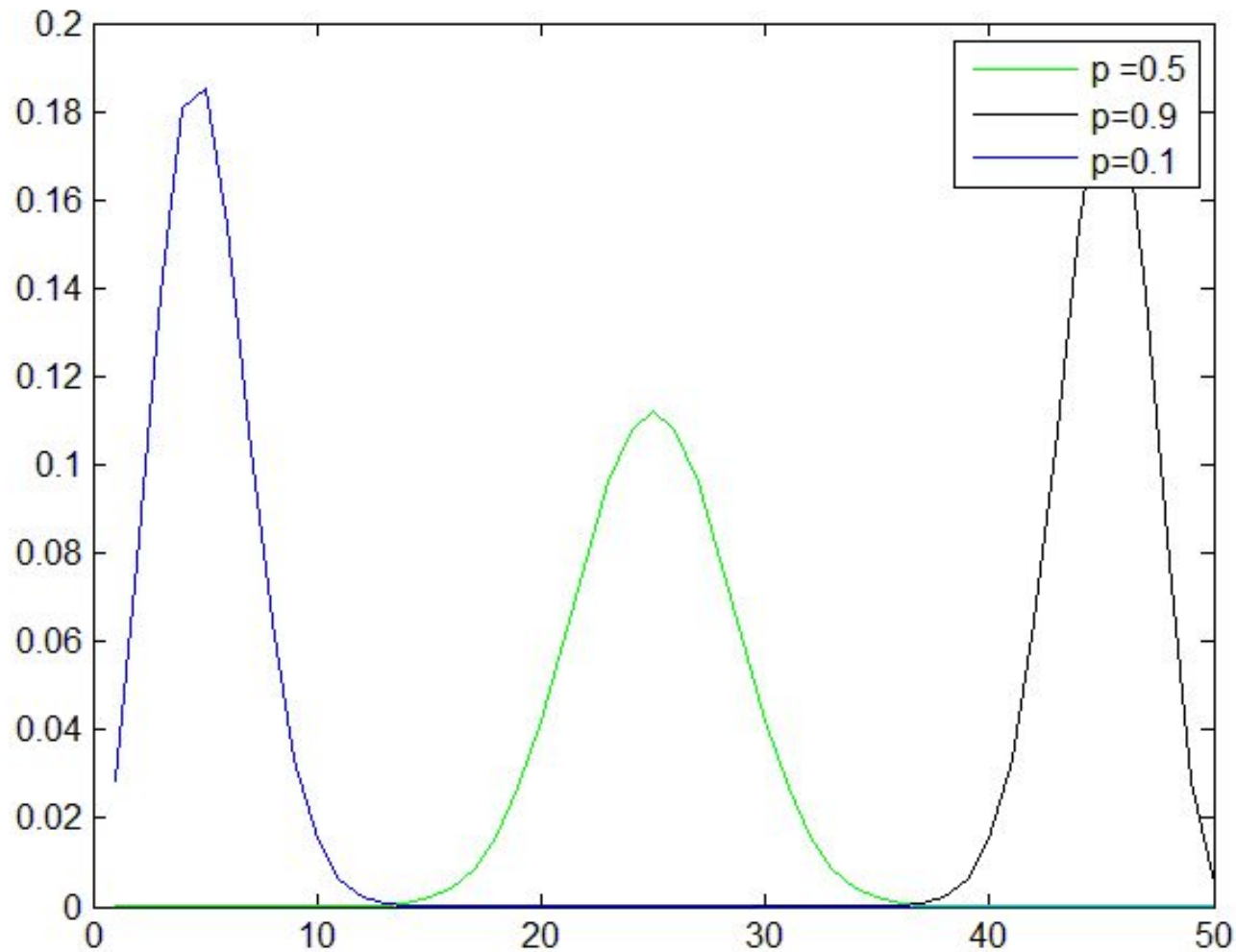
$$= \frac{1}{x\sqrt{2\pi}} \int_x^{\infty} t e^{-t^2/2} dt = \frac{e^{-x^2/2}}{x\sqrt{2\pi}}$$

Using change of variables

$$\therefore P(X > x) \leq \frac{e^{-x^2/2}}{x\sqrt{2\pi}}$$

The upper bound values become smaller and smaller as x increases

Binomial distribution and Gaussian distribution



Binomial distribution and Gaussian distribution

- The binomial distribution begins to resemble a Gaussian distribution with an appropriate mean for large values of n .
- In fact this resemblance begins to show up for surprisingly small values of n .
- Recall that a binomial random variable is the number of successes of independent Bernoulli trials

$$X = \sum_{i=1}^n X_i, X_i = 1 \text{ (heads on } i^{\text{th}} \text{ trial) else } 0$$

Binomial distribution and Gaussian distribution

- Each X_i has a mean of p and standard deviation of $p(1-p)$.
- Hence the following random variable is a standard normal random variable by CLT:

$$\frac{X - np}{\sqrt{np(1-p)}}$$

- Watch the animation [here](#).

Binomial distribution and Gaussian distribution

- Another way of stating the afore-mentioned facts is that:

When $n \rightarrow \infty$, we have $\forall a, b, a \leq b$,

$$P\left(a \leq \frac{X - np}{\sqrt{np(1-p)}} \leq b\right) = \Phi(b) - \Phi(a)$$

where $X \sim \text{Binomial}(n, p)$

- This is called the **de Moivre-Laplace theorem** and is a special case of the CLT. But its proof was published almost 80 years before that of the CLT!

Distribution of the sample mean

- Consider independent and identically distributed random variables $X_1, X_2, \dots, X_n$ with mean μ and standard deviation σ .
- We know that the sample mean (or empirical mean) is a **random variable** given by:

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

Note yet again: The true mean μ is NOT a random variable. The sample mean is, and its value converges to the true mean μ by the law of large numbers.

Distribution of the sample mean

- Now we have:

$$E(\bar{X}) = E\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \mu$$

$$Var(\bar{X}) = \frac{\sum_{i=1}^n Var(X_i)}{n^2} = \frac{\sigma^2}{n}$$

If $X_1, X_2, \dots, X_n$ were normal random variables, then it can be proved that $\bar{X}$ is also a normal random variable (how?). Otherwise if $X_1, X_2, \dots, X_n$ weren't normal random variables, $\bar{X}$ would be only *approximately* normally distributed, as per the central limit theorem.

Distribution of the sample variance

- The sample variance is given by:

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1} = \frac{\sum_{i=1}^n X_i^2 - n\bar{X}^2}{n-1}$$

- The sample standard deviation is S .

Distribution of the sample variance

- The expected value of the sample variance is derived as follows:

$$\begin{aligned} E((n-1)S^2) &= E\left(\sum_{i=1}^n X_i^2 - n\bar{X}^2\right) \\ &= nE(X_1^2) - nE(\bar{X}^2) \end{aligned}$$

$$E(W^2) = Var(W) + (E(W))^2$$

$$\begin{aligned} \therefore E((n-1)S^2) &= nVar(X_1) + n(E(X_1))^2 \\ &\quad - nVar(\bar{X}) - n(E(\bar{X}))^2 \end{aligned}$$

Distribution of the sample variance

- The expected value of the sample variance is derived as follows:

$$\begin{aligned}\therefore E((n-1)S^2) &= n\text{Var}(X_1) + n(E(X_1))^2 \\ &\quad - n\text{Var}(\bar{X}) - n(E(\bar{X}))^2\end{aligned}$$

$$\begin{aligned}\therefore E((n-1)S^2) &= n\sigma^2 + n\mu^2 \\ &\quad - n(\sigma^2 / n) - n(\mu)^2 = (n-1)\sigma^2\end{aligned}$$

$$\therefore E(S^2) = \sigma^2$$

Distribution of the sample variance

- The expected value of the sample variance is derived as follows:

$$\therefore E((n-1)S^2) = (n-1)\sigma^2$$

$$\therefore E(S^2) = \sigma^2$$

If the sample variance were instead defined as

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n} = \frac{\sum_{i=1}^n X_i^2 - n\bar{X}^2}{n},$$

we would have :

$$E(S^2) = \frac{(n-1)\sigma^2}{n}$$

This is **undesirable** – as we would like to have the expected value of the sample variance to equal the true variance! Hence S^2 here above is multiplied by $(n-1)/n$ to correct for this anomaly giving rise to our strange definition of sample variance. This multiplication by $(n-1)/n$ is called **Bessel's correction**.

Distribution of the sample variance

- But the mean and the variance alone does not determine the distribution of a random variable.
- So what about the distribution of the sample variance?
- For that we need to study another distribution first – the **chi-squared distribution**.

Chi-square distribution

- If $Z_1, Z_2, \dots, Z_n$ are *independent* standard normal random variables, then the following quantity

$$X = Z_1^2 + Z_2^2 + \dots + Z_n^2$$

is said to have a chi-square distribution with n degrees of freedom and is denoted as follows

$$X \sim \chi_n^2$$

- The formula for this is as follows:

$$f_X(x) = \frac{x^{n/2-1} e^{-x/2}}{2^{n/2} \Gamma(n/2)}$$

$$\Gamma(y) = (y-1)! \quad (y \text{ integer})$$

$$= \int_0^{\infty} x^{y-1} e^{-x} dx \quad (y \text{ real})$$

Chi-square distribution

- To obtain the expression for the chi-square distribution when $n = 1$:

$$Z_1 \sim N(0,1)$$

$$X = Z_1^2$$

$$\begin{aligned} F_X(x) &= P(Z_1^2 \leq x) = P(-\sqrt{x} \leq Z_1 \leq \sqrt{x}) \\ &= F_{Z_1}(\sqrt{x}) - F_{Z_1}(-\sqrt{x}) \end{aligned}$$

$$\begin{aligned} f_X(x) &= \frac{1}{2\sqrt{x}} f_{Z_1}(\sqrt{x}) + \frac{1}{2\sqrt{x}} f_{Z_1}(-\sqrt{x}) \\ &= \frac{1}{2\sqrt{x}} \frac{e^{-x/2}}{\sqrt{2\pi}} \times 2 = \frac{1}{\sqrt{x}} \frac{e^{-x/2}}{\sqrt{2\pi}} \end{aligned}$$

The corresponding derivation for arbitrary n is quite complicated, and relies on geometrical properties of spheres in n dimensional space. See [here](#) (not on exam).

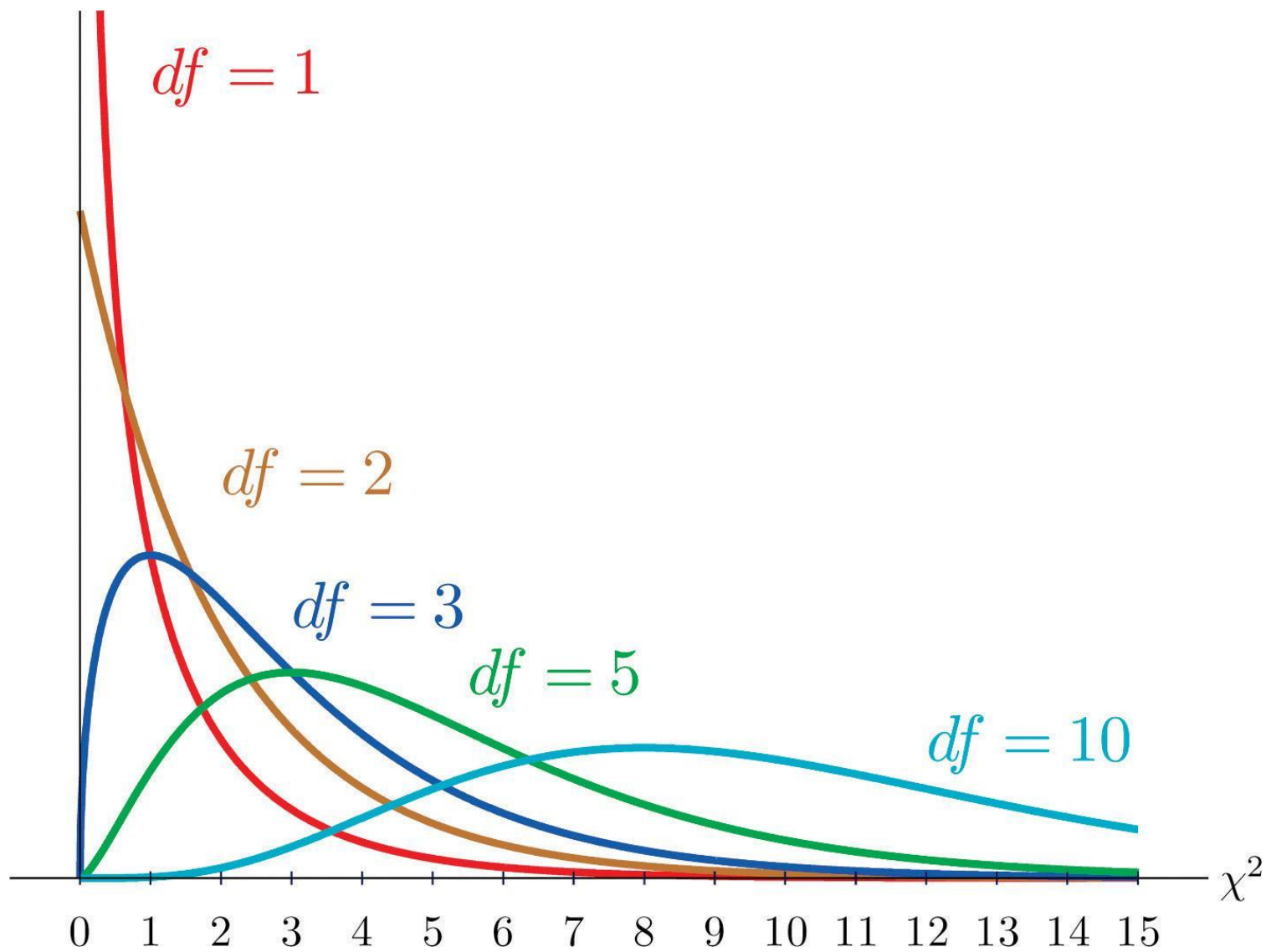
Chi-square distribution

- MGF of a chi-square distribution with n degrees of freedom:

$$\Phi_X(t) = (1 - 2t)^{-n/2}$$

Proof on the board. And here.

- Please note that the aforementioned MGF is defined only for $t < 1/2$.



Additive property

- If X_1 and X_2 are independent chi-square random variables with n_1 and n_2 degrees of freedom respectively, then $X_1 + X_2$ is also a chi-square random variable with $n_1 + n_2$ degrees of freedom. This is called the **additive property**.
- It is easy to prove this property by observing that $X_1 + X_2$ is basically the sum of $n_1 + n_2$ independent normal random variables.

Chi-square distribution

- Tables for the chi-square distribution are available for different number of degrees of freedom, and for different values of the independent variable.

Back to the distribution of the sample variance

We will assume that the samples $X_1, \dots, X_n$ are independently drawn from $N(\mu, \sigma^2)$

$$(n-1)S^2 = \sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2$$

$$\therefore \frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} + \frac{n(\bar{X} - \mu)^2}{\sigma^2}$$

$$\therefore \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} + \left(\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \right)^2$$

The sum of squares of n standard normal random variables

The square of a standard normal random variable

Back to the distribution of the sample variance

We will assume that the samples $X_1, \dots, X_n$ are independently drawn from $N(\mu, \sigma^2)$

$$\therefore \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} + \left(\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \right)^2$$

The sum of squares of n standard normal random variables

The square of a standard normal random variable

It turns out that these two quantities are **independent** random variables. The proof of this requires multivariate statistics and transformation of random variables, and is deferred to a later point in the course. If you are curious, you can browse [this link](#), but it's not on the exam for now.

Given this fact about independence, it then follows that the middle term is a chi-square distribution with $n-1$ degrees of freedom.

Back to the distribution of the sample variance

- The previous two slides assumed that the samples $X_1, \dots, X_n$ were independently drawn from $N(\mu, \sigma^2)$.
- Consider the case that the samples were not Gaussian, but all the same **finite** mean μ and the same **finite** variance σ^2 .
- In such cases, the RHS term involving $\bar{X}$ is still **approximately** standard normal due to the CLT.
- Likewise, the sum on the LHS is also **approximately** standard normal due to the CLT.

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} + \left(\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \right)^2$$

The sum of squares of n random variables that are approximately distributed as $N(0,1)$ – due to the CLT

The square of a random variable that is approx. from $N(0,1)$ – due to the CLT

Uniform distribution

Uniform distribution

- A uniform random variable over the interval $[a,b]$ has a pdf given by:

$$f_X(x) = 1/(b-a), \text{ if } a < x < b \\ = 0 \text{ otherwise}$$

- Clearly, this is a valid pdf – it is non-negative and integrates to 1.
- It is easy to show that its mean and median are equal to $(b+a)/2$.

$$E(X) = \int_a^b \frac{xdx}{b-a} = \frac{x^2 \big|_a^b}{2(b-a)} = \frac{a+b}{2}$$

Uniform distribution

- Variance:

$$E(X^2) = \int_a^b \frac{x^2 dx}{b-a} = \frac{x^3 \Big|_a^b}{3(b-a)} = \frac{b^3 - a^3}{3(b-a)} = \frac{b^2 + a^2 + ab}{3}$$

$$Var(X) = E(X^2) - (E(X))^2 = \frac{(b-a)^2}{12}$$

- MGF:

$$E(e^{tX}) = \int_a^b \frac{e^{tx} dx}{b-a} = \frac{e^{tb} - e^{ta}}{t(b-a)}, t \neq 0$$

$$= 1, t = 0$$

Applications

- Uniform random variables, especially over the $[0,1]$ interval are very important, in developing programs to draw samples from other distributions including the Gaussian, Poisson, and others.
- For now, we will study two applications. How do you draw a sample from a distribution of the following form:

$$P(X = x_i) = p_i, 1 \leq i \leq n,$$

$$\sum_{i=1}^n p_i = 1$$

Applications

- For now, we will study two applications. How do you draw a sample from a discrete distribution with the following pmf:

$$P(X = x_i) = p_i, 1 \leq i \leq n, \sum_{i=1}^n p_i = 1$$

Draw $u \sim \text{Uniform}(0,1)$

If $u \leq p_1 \rightarrow$ sampled value is x_1

If $p_1 \leq u \leq p_1 + p_2 \rightarrow$ sampled value is x_2

.

.

If $p_1 + p_2 + \dots + p_{n-1} \leq u \leq p_1 + p_2 + \dots + p_{n-1} + p_n \rightarrow$ sampled value is x_n

Applications

- Uniform random variables, especially over the $[0,1]$ interval are very important, in developing programs to draw samples from other distributions including the Gaussian, Poisson, and others.
- For now, we will study how to generate a random permutation of n elements. That is what the “**randperm**” function in MATLAB does, and you have used it at least once so far!

Application: generating a random subset

- In fact, we will do something more than randperm – we will develop theory to generate a random subset of size k from a set $A = \{a_1, a_2, \dots, a_n\}$ of size n , assuming all the $C(n, k)$ subsets are equally likely.

- Let us define the following for each element j ($1 \leq j \leq n$):

$$I_j = 1 \text{ if } a_j \in B_k, \text{ else } 0$$

Notation for chosen subset
(of size k)

- Now we will sequentially pick each element of the subset randomly as follows:

Application: generating a random subset

- Notice that $P(I_1=1) = k/n$. (why? Because there are $C(n,k)$ ways to pick k objects out of n . Let us say object 1 is one of them, then there are $C(n-1,k-1)$ ways to pick the other $k-1$ objects)
- If I_1 is 1, then $P(I_2=1) = (k-1)/(n-1)$. (why?)
- If I_1 is 0, then $P(I_2=1) = k/(n-1)$. (why?)
- Thus $P(I_2=1|I_1) = (k-I_1)/(n-1)$ (why?)
- Side question: what is $P(I_2=1)$?

Application: generating a random subset

- Continuing this way, one can show that:

$$P(I_j \mid I_1, I_2, \dots, I_{j-1}) = \frac{k - \sum_{i=1}^{j-1} I_i}{n - (j-1)}, 2 \leq j \leq n$$

Application: generating a random subset

- This suggests the following procedure:

$$U_1 \sim \text{Uniform}(0,1)$$

$$I_1 = 1, \text{ if } U_1 < k/n, \text{ else } 0$$

$$U_2 \sim \text{Uniform}(0,1)$$

$$I_2 = 1, \text{ if } U_2 < (k - I_1)/(n - 1), \text{ else } 0$$

.

.

$$U_j \sim \text{Uniform}(0,1)$$

$$I_j = 1, \text{ if } U_j < (k - I_1 - I_2 - \dots - I_{j-1})/(n - j + 1), \text{ else } 0$$

When does this process stop? It stops at step # j

* If $I_1 + I_2 + \dots + I_j = k$ and the random subset B_k contains those indices whose I -values are 1

OR

* If the number of unfilled entries in the random subset B_k = number of remaining elements in A . In this case B_k = all remaining elements in A with index greater than i = largest index in B_k . See figure 5.6 of the book.

Sampling from arbitrary distributions

- If X has distribution $F_X(\cdot)$, then the random variable $Y = F_X(X)$ has a $[0,1]$ uniform distribution.
- The proof of this can be found on [this link](#).
- This is true for any distribution of X .
- This fact can be used to generate samples from any arbitrary distribution:
 - Generate a sample value z from $\text{Uniform}(0,1)$
 - Then $(F_X)^{-1}(z)$ is a sample from F_X .
 - Proof: next slide

$$z \sim \text{Uniform}(0, 1)$$

$$P(F_X^{-1}(z) \leq q) = P(z \leq F_X(q))$$

$$= F_X(q) \text{ as } z \text{ is a uniform r.v. and hence } P(z \leq z') = z'$$

$$\therefore F_X^{-1}(z) \text{ has distribution } \text{Uniform}(0, 1)$$

Exponential Distribution

Motivation

- Consider a Poisson distribution with an average number of successes per unit time given by λ .
- So the number of successes in time u is λu .
- This is actually called a **Poisson process**.
- Now consider the time taken (T) for the first success – this is called as the **waiting time**.

Motivation

- Let $X \sim \text{Poisson}(\lambda u)$ for time interval $(0, u)$.
- T is a random variable whose distribution we are going to seek to model here. Then,

$$P(T \leq u) = 1 - P(T > u) = 1 - P(X = 0)$$

The probability that the first success occurred after time u = probability that there was no success in the time interval $(0, u)$, i.e. $X = 0$ in that interval.

$$\begin{aligned} P(T \leq u) &= 1 - P(T > u) = 1 - P(X = 0) \\ &= 1 - \frac{e^{-\lambda u} (\lambda u)^0}{0!} = 1 - e^{-\lambda u} \end{aligned}$$

Motivation

$$F_T(u) = 1 - e^{-\lambda u}$$

$$f_T(u) = \lambda e^{-\lambda u}, u \geq 0$$

(0 elsewhere)

Such a random variable T is called an **exponential random variable**. It models the waiting time for a Poisson process. It has a parameter λ .

Properties: MGF

- MGF defined for $t < \lambda$:

$$\Phi_T(t) = E(e^{tT})$$

$$= \int_0^{\infty} e^{tu} \lambda e^{-\lambda u} du = \lambda \int_0^{\infty} e^{-(\lambda-t)u} du = \frac{\lambda}{\lambda - t}$$

Properties: Mean and Variance

- Mean:

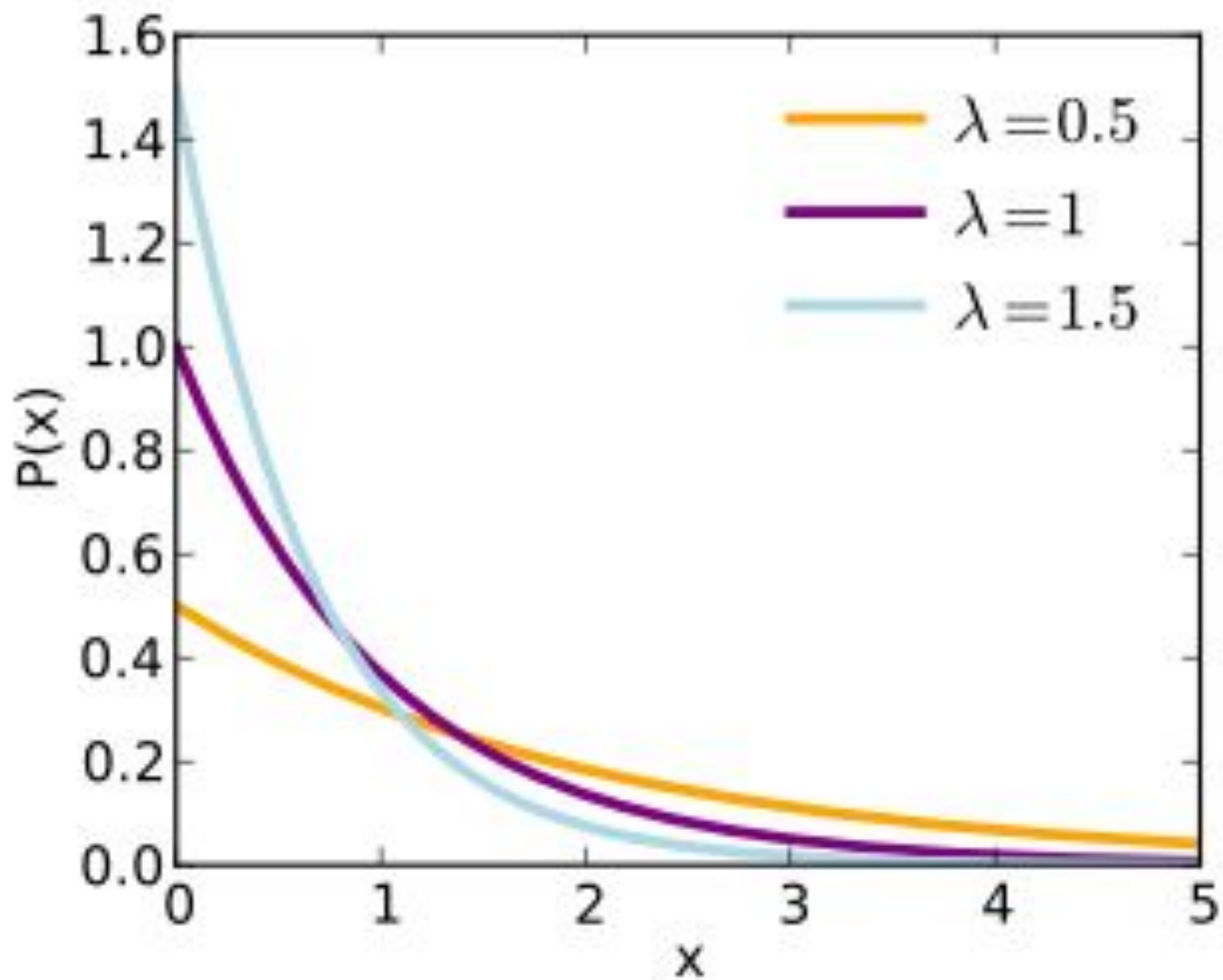
$$E(T) = \int_0^{\infty} t\lambda e^{-\lambda t} dt = \lambda \int_0^{\infty} te^{-\lambda t} dt = \frac{1}{\lambda}$$

This is intuitive – a Poisson process with a large average rate should definitely lead to a lower expected waiting time.

- Variance:

$$Var(T) = E(T^2) - (E(T))^2 = \frac{1}{\lambda^2}$$

$$E(T^2) = \int_0^{\infty} t^2 \lambda e^{-\lambda t} dt = \frac{2}{\lambda^2}$$



Properties: Mode and Median

- Mode: always at 0

- Median

$$\int_0^u \lambda e^{-\lambda t} dt = \frac{1}{2}$$

$$u = \frac{\ln 2}{\lambda}$$

Properties: “Memorylessness”

- A non-negative random variable T is said to be **memoryless** if:

$$\forall s, u \geq 0, P(T > s + u \mid T > u) = P(T > s)$$

- Meaning: This gives the probability that given a waiting time of success of more than u , the waiting time will exceed $s+u$, i.e. one would have to wait for s more time units for success.
- Another formula (equivalent to the earlier one)

$$\frac{P(T > s + u, T > u)}{P(T > u)} = P(T > s)$$

Properties: “Memorylessness”

- You can easily verify that this holds for the exponential distribution.

$$P(T > u) = e^{-\lambda u}$$

$$P(T > s) = e^{-\lambda s}$$

$$P(T > s + u) = e^{-\lambda(s+u)}$$

Example

- Suppose that the number of miles a car can run before its battery fails is exponentially distributed with an average of α . What is the probability that the car won't fail on a trip of k miles given that it has already run for l miles?
- Solution: For exponential distribution we know that

$$P(T > k + l \mid T > l) = \frac{P(T > k + l, T > l)}{P(T > l)}$$

$$= P(T > k) = e^{-k\lambda} = e^{-k/\alpha}$$

Example

- Suppose that the number of miles a car can run before its battery fails is exponentially distributed with an average of α . What is the probability that the car won't fail on a trip of k miles given that it has already run for l miles?
- Solution: If the distribution were not exponential, then we have

$$P(T > k + l \mid T > l) = \frac{P(T > k + l, T > l)}{P(T > l)} = \frac{1 - F_T(k + l)}{1 - F_T(l)}$$

Property: Minimum

- Consider independent exponentially distributed random variables $X_1, X_2, \dots, X_n$ with parameters $\lambda_1, \lambda_2, \dots, \lambda_n$. Then $\min(X_1, X_2, \dots, X_n)$ is exponentially distributed.

$$P(\min(X_1, X_2, \dots, X_n) > x) = P(X_1 > x, X_2 > x, \dots, X_n > x)$$

$$= \prod_{i=1}^n P(X_i > x) \text{ due to independence}$$

$$= \prod_{i=1}^n e^{-\lambda_i x}$$

$$= e^{-\sum_{i=1}^n \lambda_i x}$$